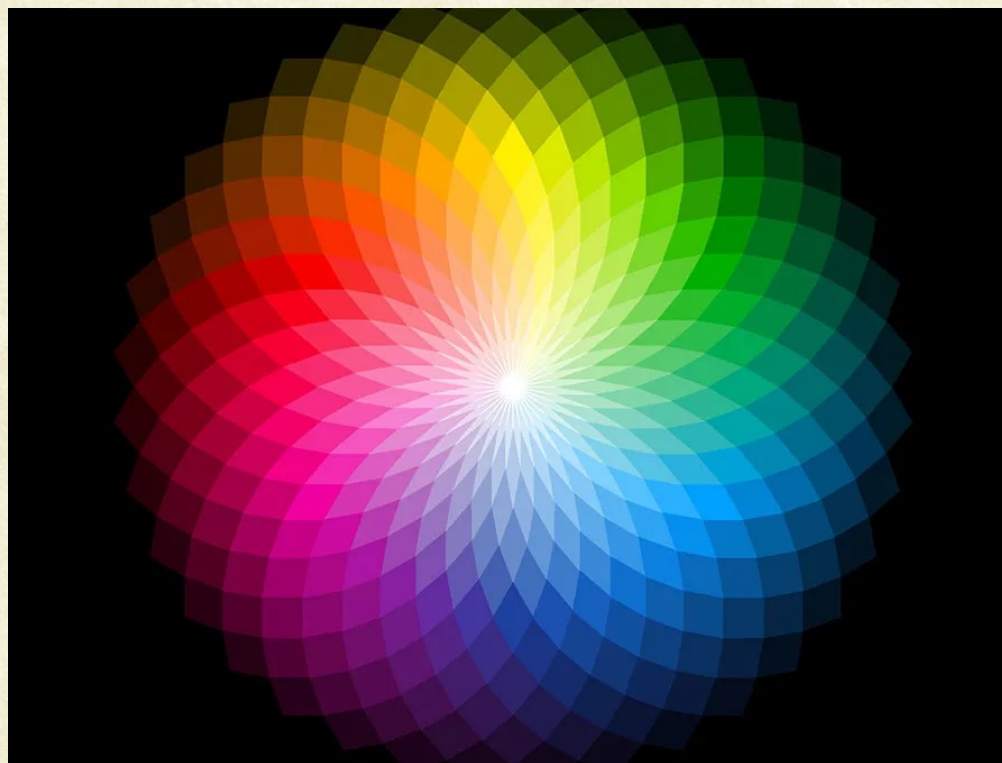




CS7.404: Digital Image Processing

Monsoon 2025: Color in Images



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



What is Color?

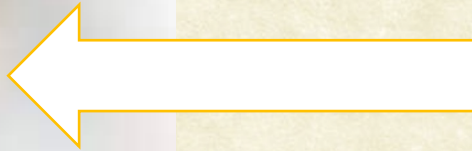
- Physical:
- Physio-
psychological:





Color as a Physical Phenomenon

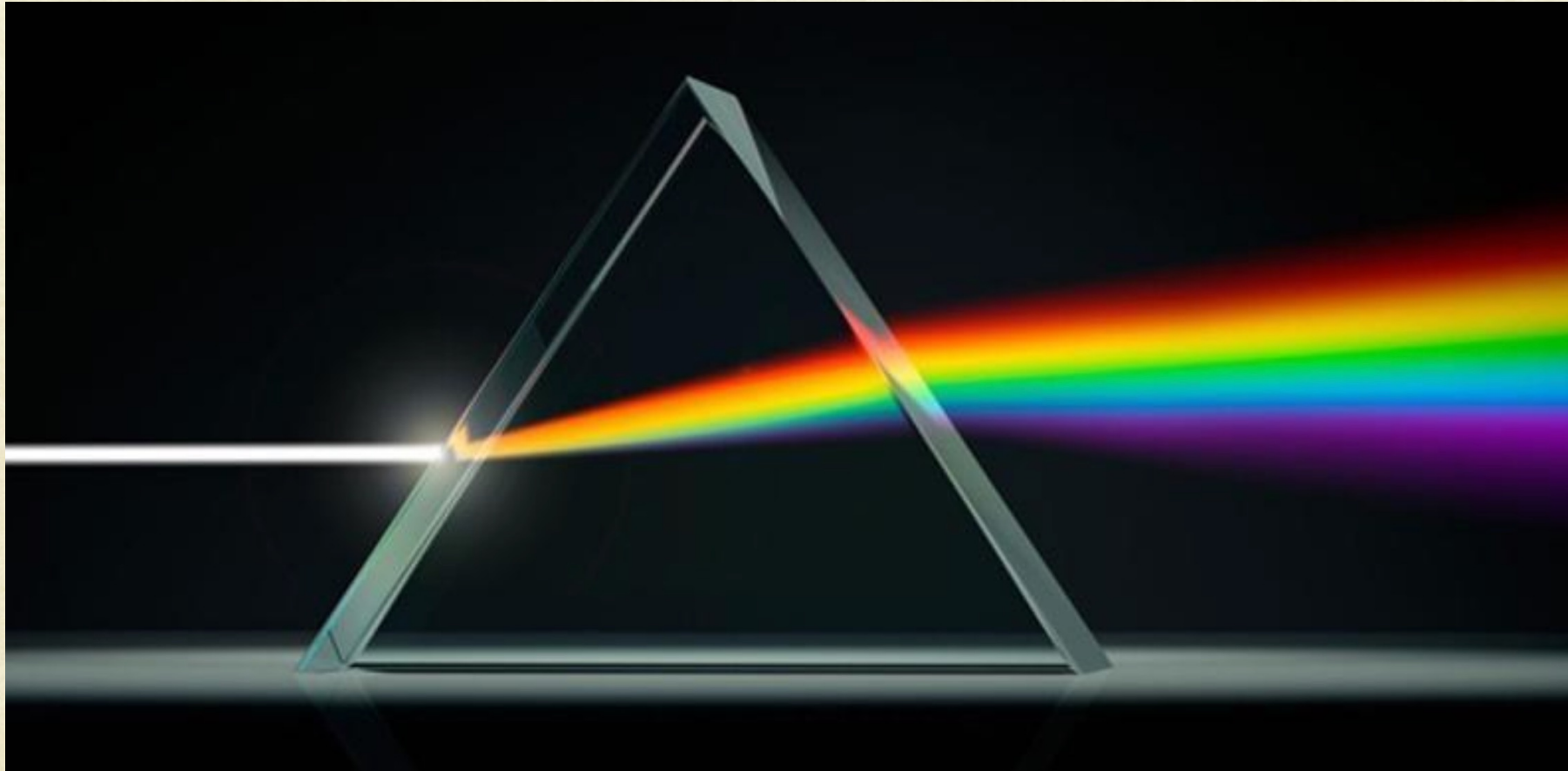
- The light falls on a surface
- The surface “reflects” it
- The resulting light falls on the eye





Color as a Physical Phenomenon: The Light

- White light is composed of EMRs of several frequencies.

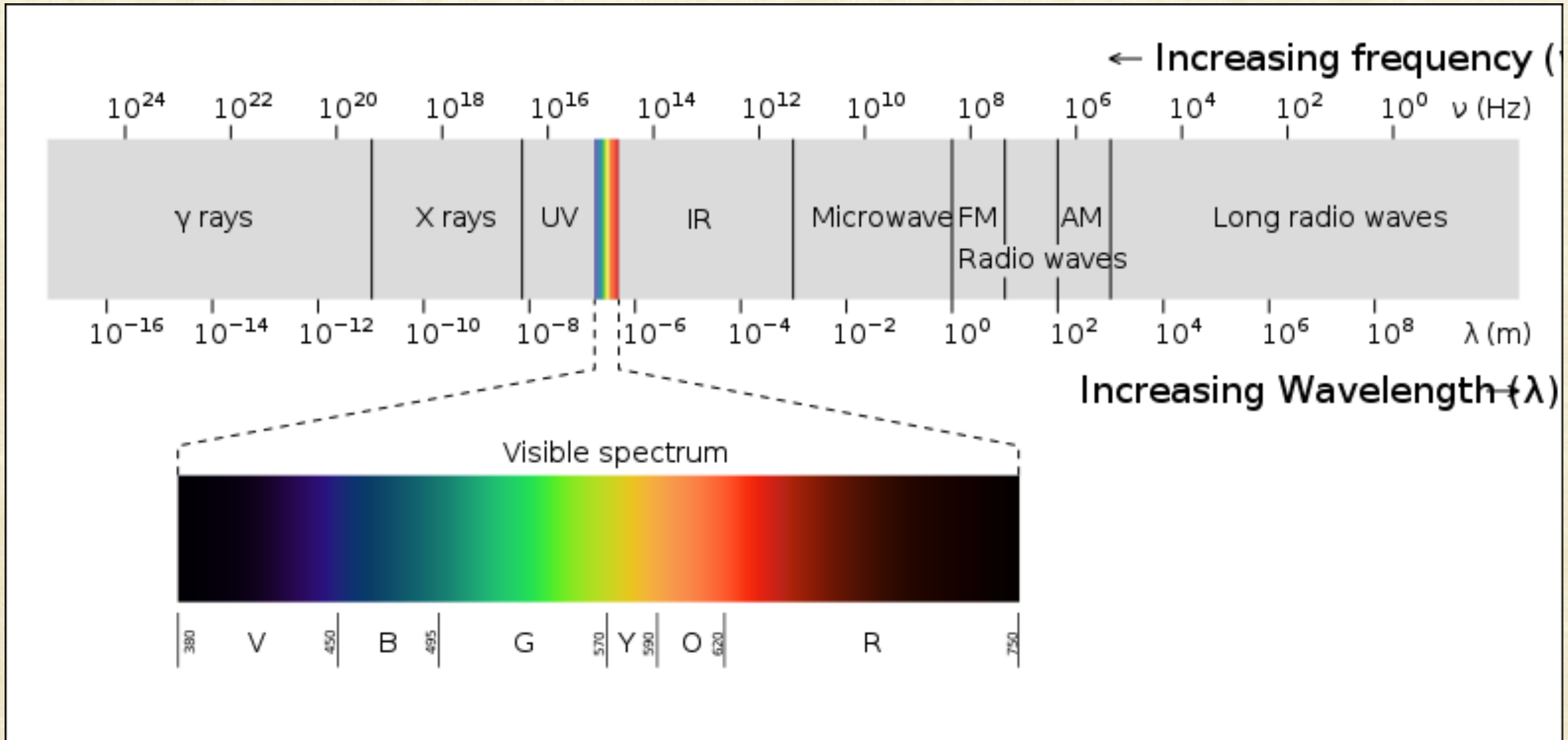


Red
Orange
Yellow
Green
Blue
Indigo
Violet

“The **rays** to speak properly are not coloured. In them there is nothing else than a certain Power and Disposition to stir up a Sensation of this or that Colour.” - Isaac Newton, Opticks, 1704



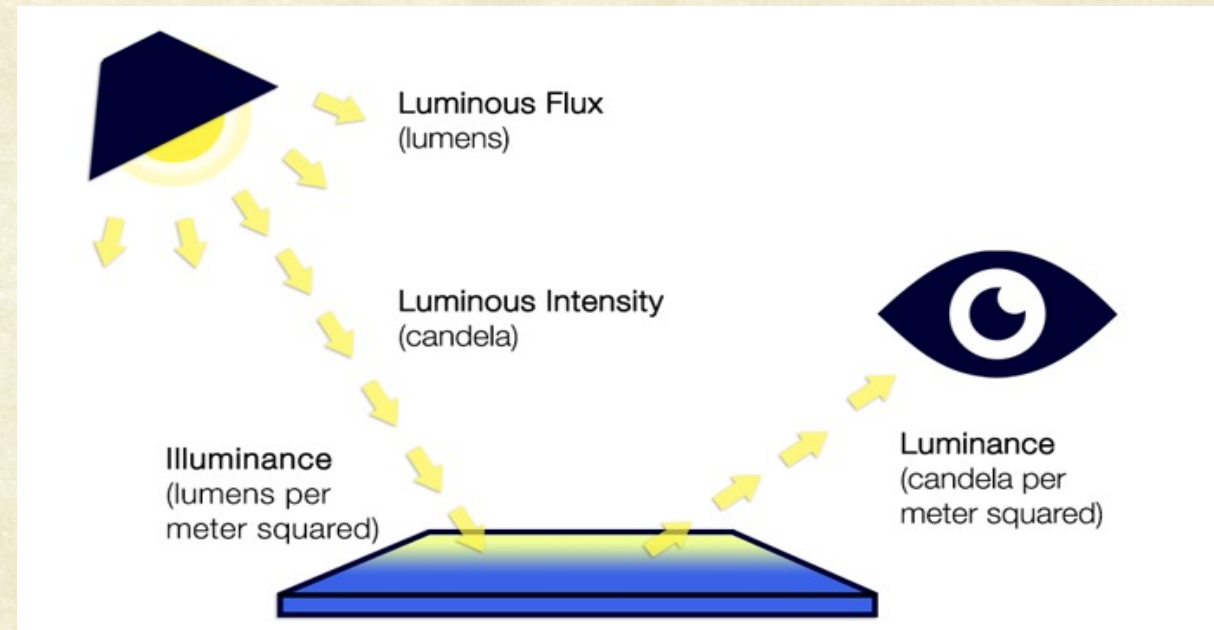
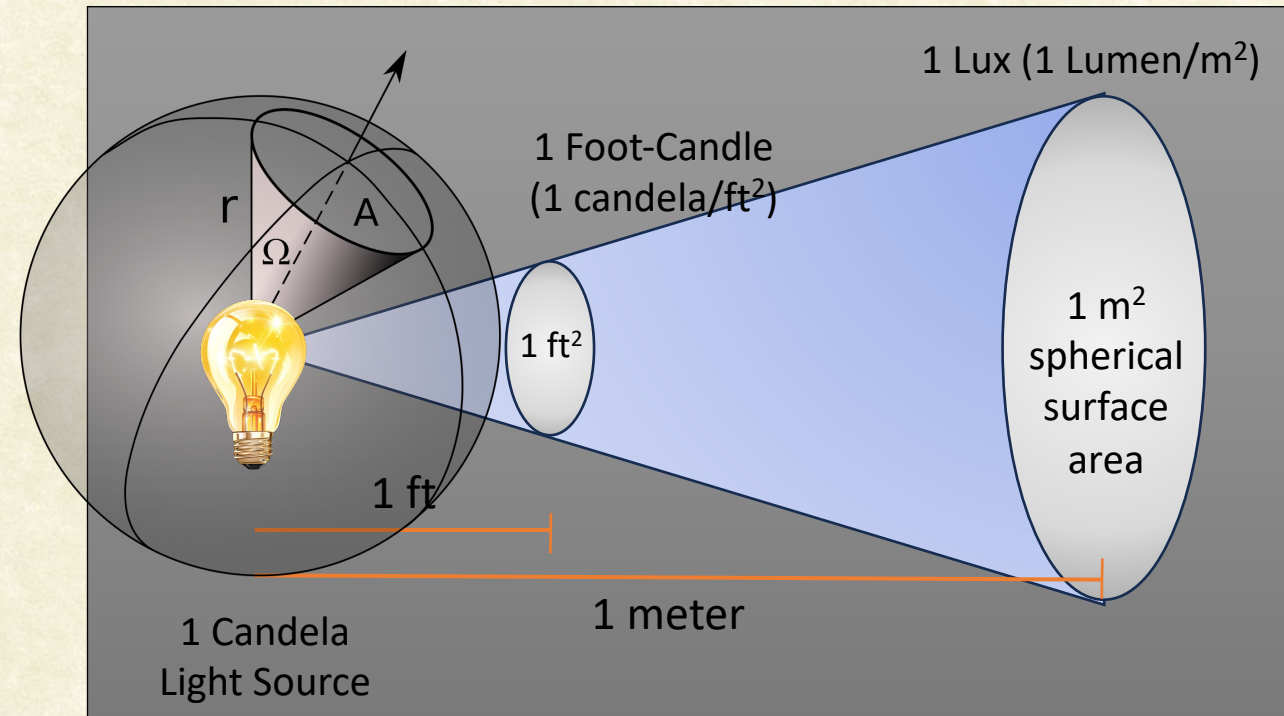
The Light





Physical Quantities

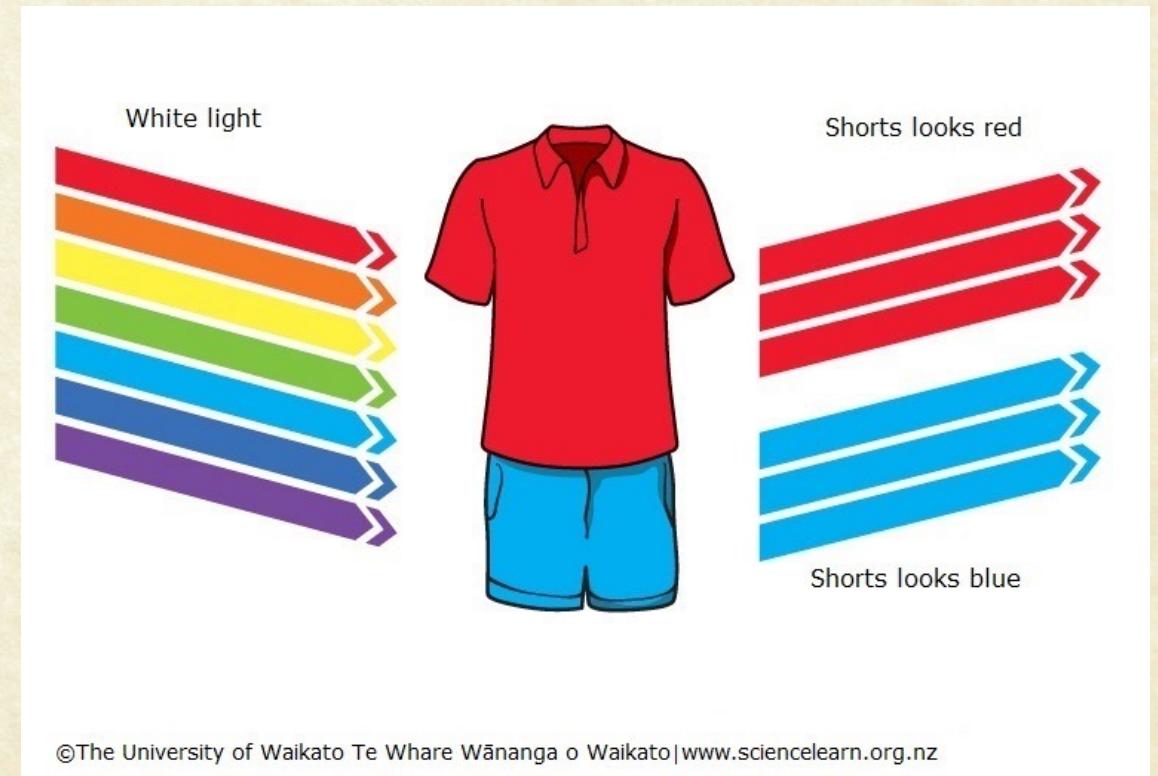
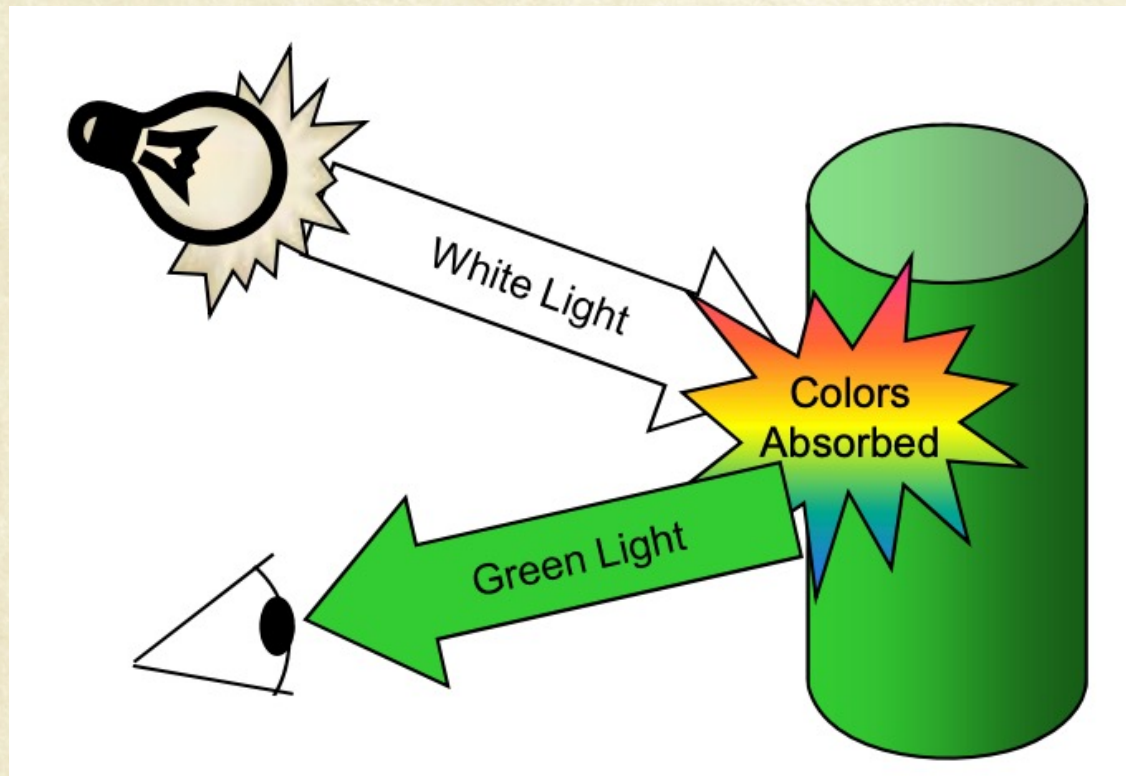
- **Radiance:** Total amount of energy flow from the light source, measured in **watts (W)**
- **Luminance:** amount of energy an observer *perceives* from a light source, measured in **candela/m²**
 - Far infrared light: high radiance, but 0 luminance
- **Brightness:** subjective descriptor that is hard to measure, (similar to the achromatic notion of intensity)





Physical Phenomenon: The Surface

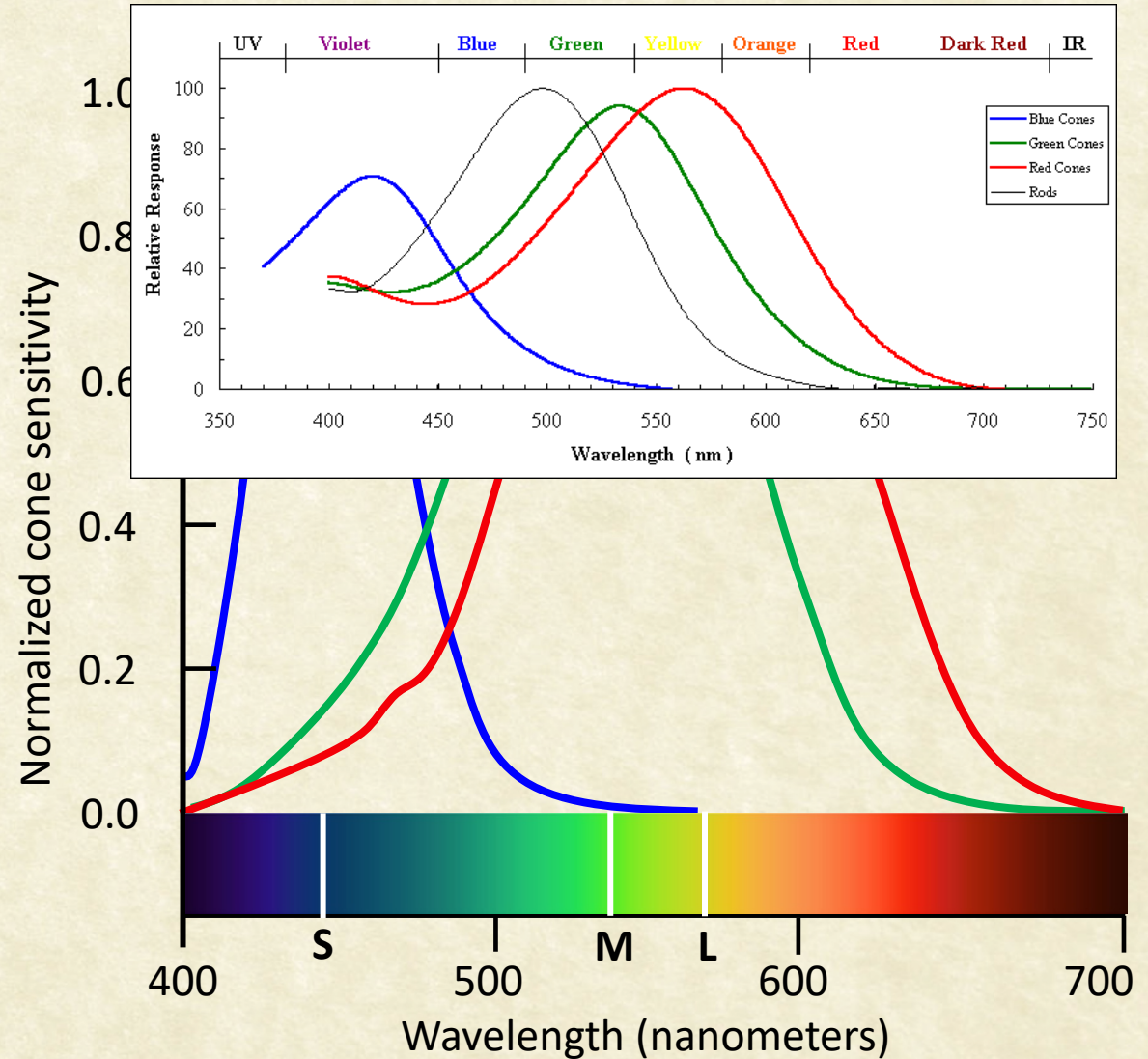
- Color is determined by nature of light **reflected** from an object
- e.g., green object reflects light from 500-570nm wavelength, and absorbs other wavelengths





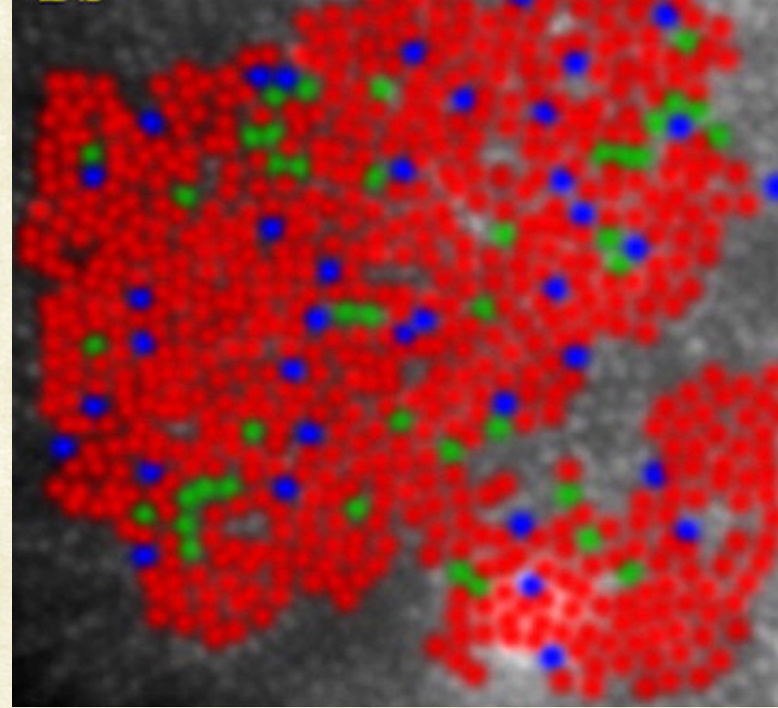
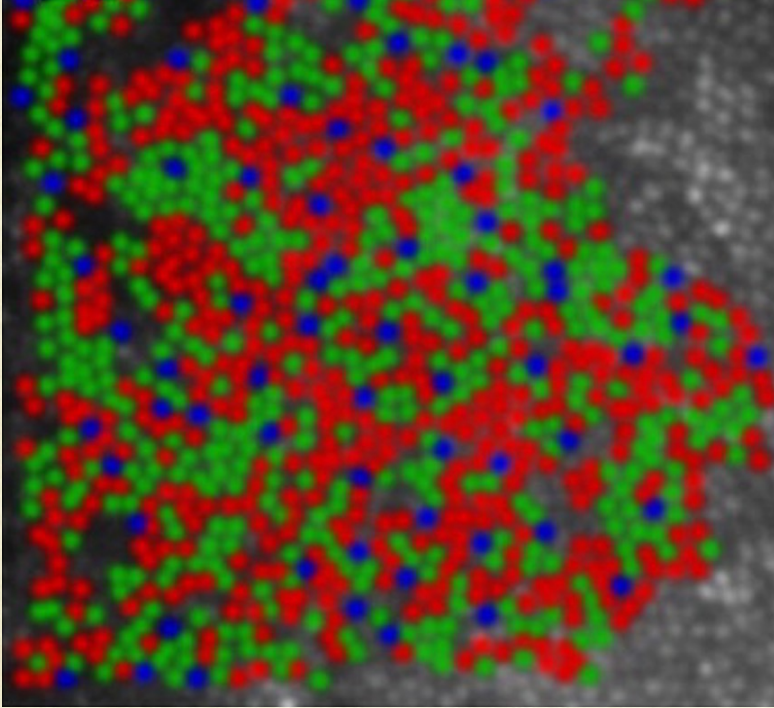
Physical Phenomenon: The Eye

- Cones are active under high light levels.
They are responsible for photopic vision.
- Three cone types are sensitive to Short (S), Medium (M) and Long (L) wavelengths.
The color names for these cones are not very accurate.





The Eye: Cone distribution

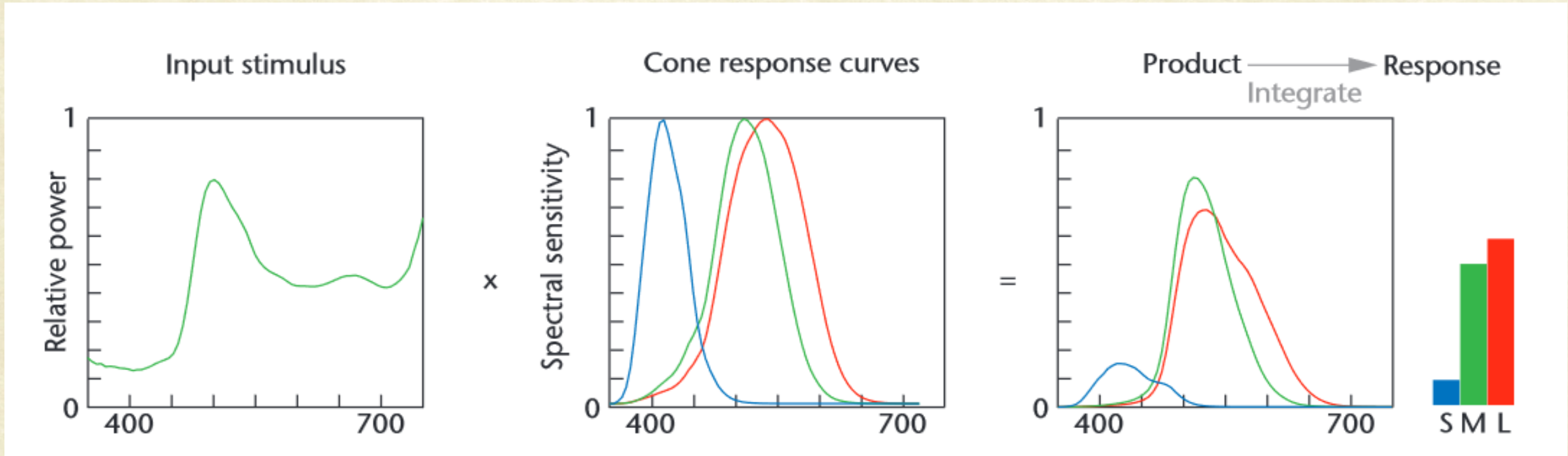


- Near-fovea images of two different human subjects: colors indicate L (red), M (green), and S (blue) cones.
- Remarkably, both subjects have normal color vision !!!

Image Courtesy: Hofer et al.



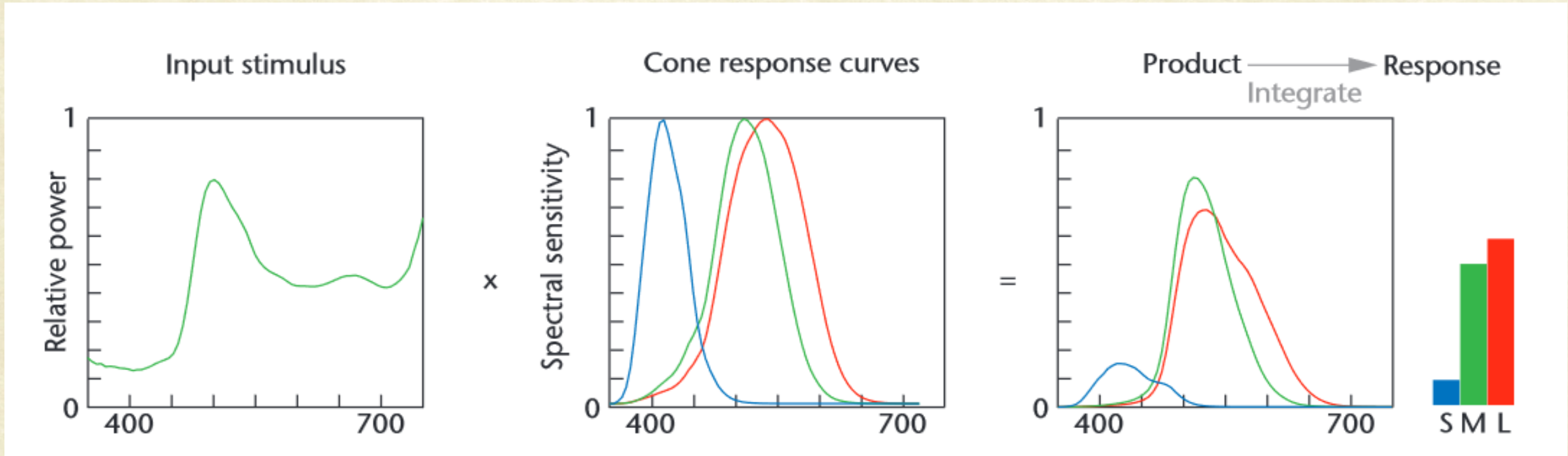
The Eye: Color signal to the brain



- Note that intensity is a weighted function of their r, g, and b values.
- The human eye does not weight each component equally.
 - $Intensity = 0.299 \times \text{Red} + 0.587 \times \text{Green} + 0.144 \times \text{Blue}.$



Psychological Phenomenon: The Brain



- The perception of color is an entirely arbitrary creation of our nervous system. It is not contained in the wavelengths themselves (remember the remark of Isaac Newton).



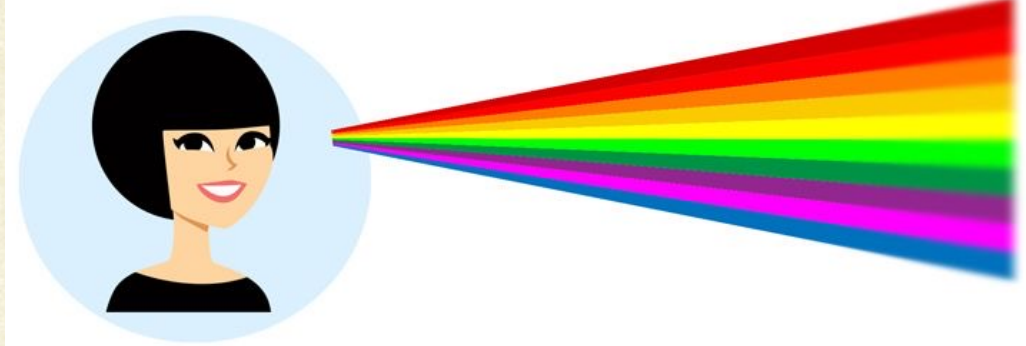
Tetrachromats:

- People with 4 different cones



*Mantis shrimp
12 primary color receptors*

SEE 10 COLORS IN THE RAINBOW
can differentiate one HUNDRED million colors



Mrs M - an English social worker,
and the first known human
"tetrachromat", discovered at
Cambridge in 1993.

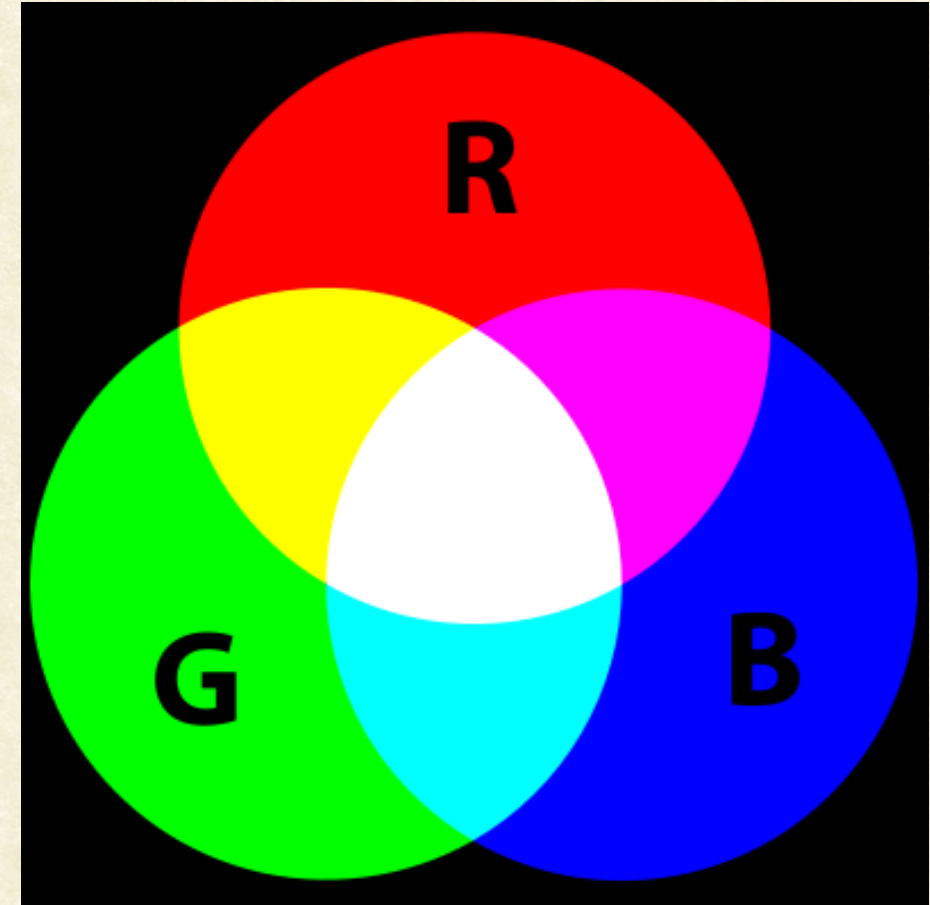
ORDINARY PEOPLE
SEE 5 COLORS IN THE RAINBOW
can differentiate one million colors





Primary Colors (color as three numbers)

- Additive (CRT displays, projectors etc.)
 - Combining red, green and blue light produces lighter colors, offering a good contrast to dark screens.
 - (Usually) Combined on a surface which reflects all light that falls on it
 - Easy to get **black** – don't shine any light (CRT)



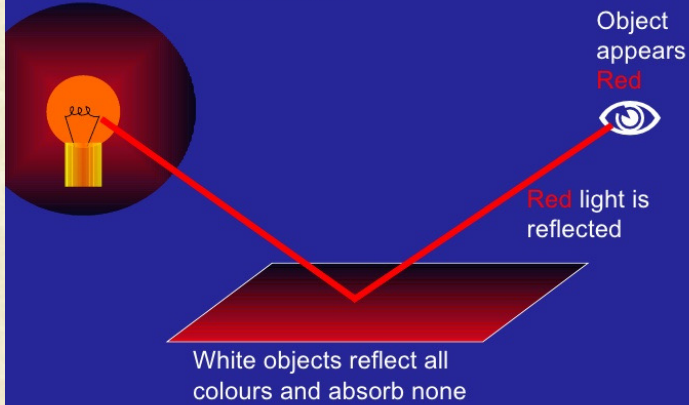
Magenta = Red + Blue
Cyan = Blue + Green
Yellow = Green + Red



Perceived Color = f(Light, Object)

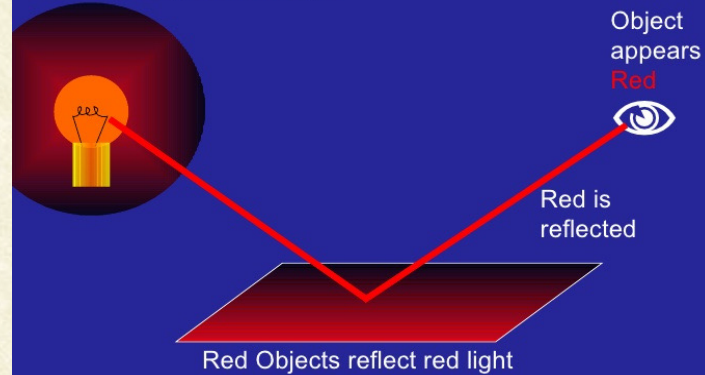
OBJECTS IN COLOURED LIGHT

White object in **red** light



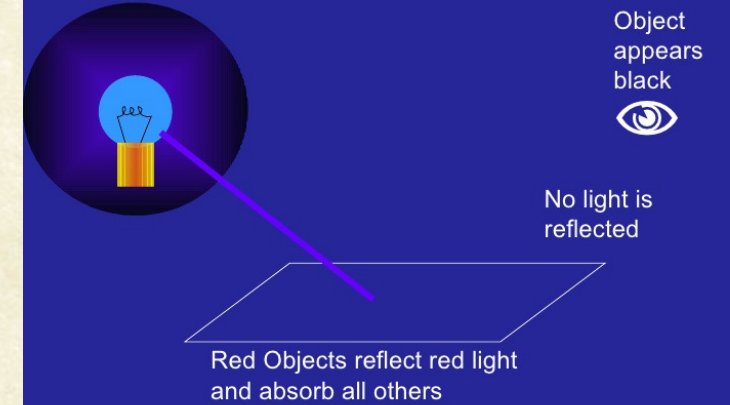
OBJECTS IN COLOURED LIGHT

Red object in **red** light

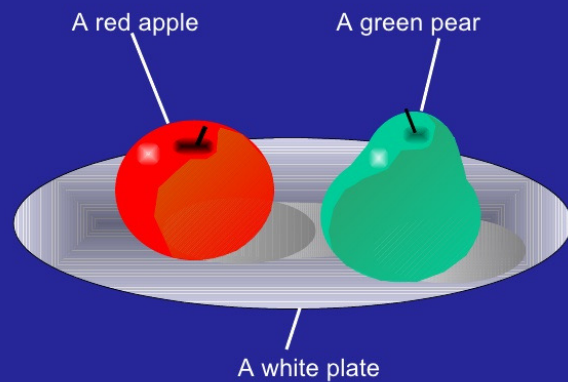


OBJECTS IN COLOURED LIGHT

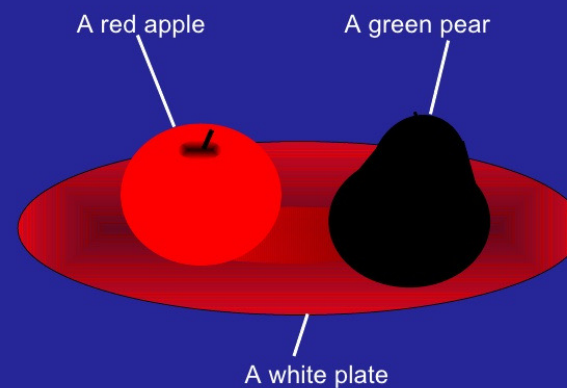
Red object in **blue** light



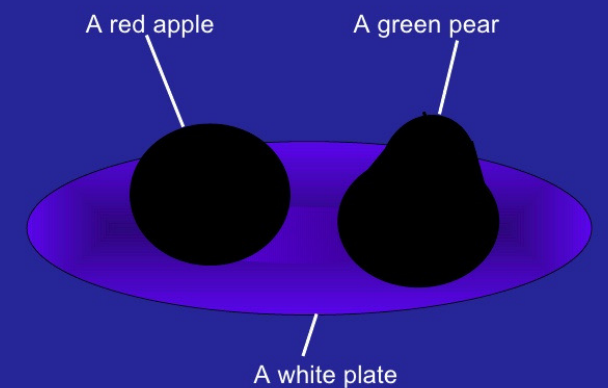
THE FRUIT BOWL IN WHITE LIGHT



THE FRUIT BOWL IN **RED** LIGHT



THE FRUIT BOWL IN **BLUE** LIGHT





R,G,B Channels of an Image



Full color



Red



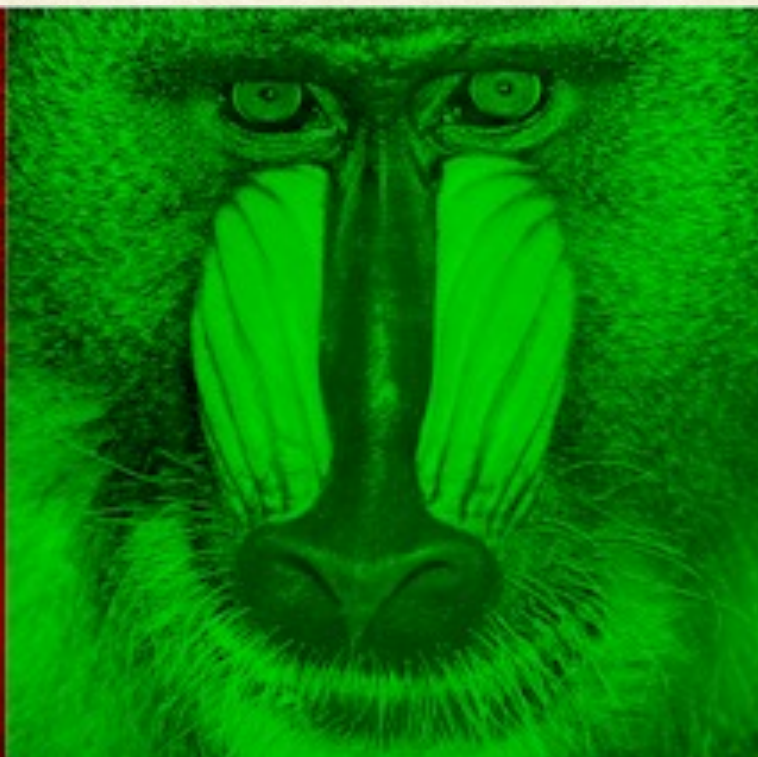
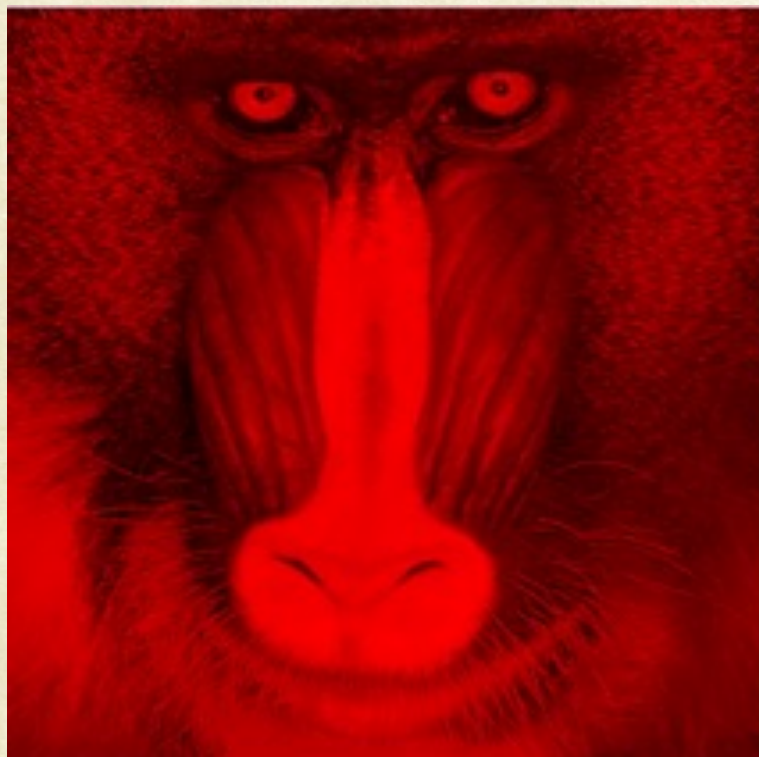
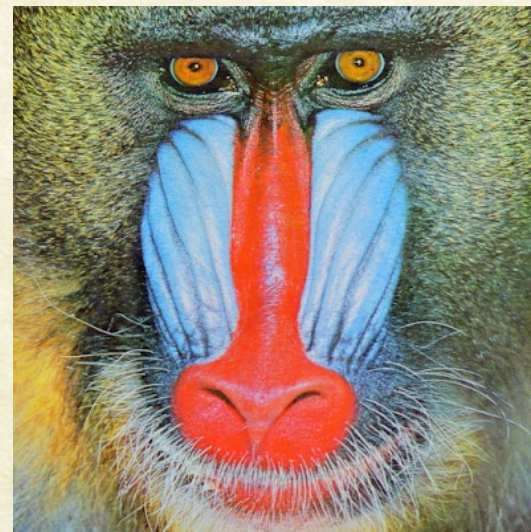
Green



Blue



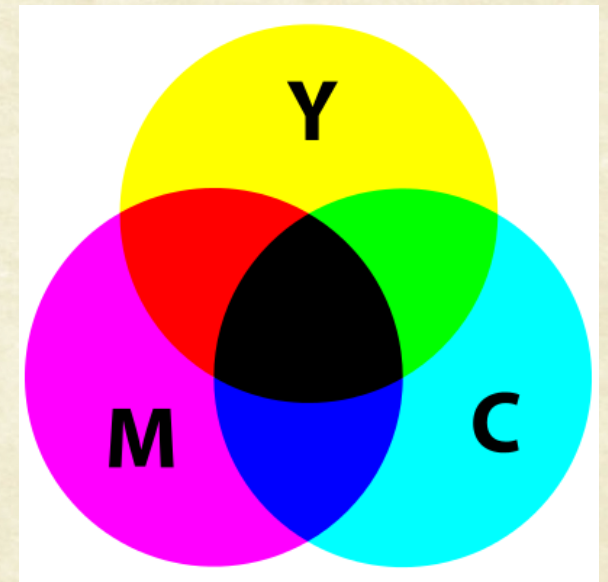
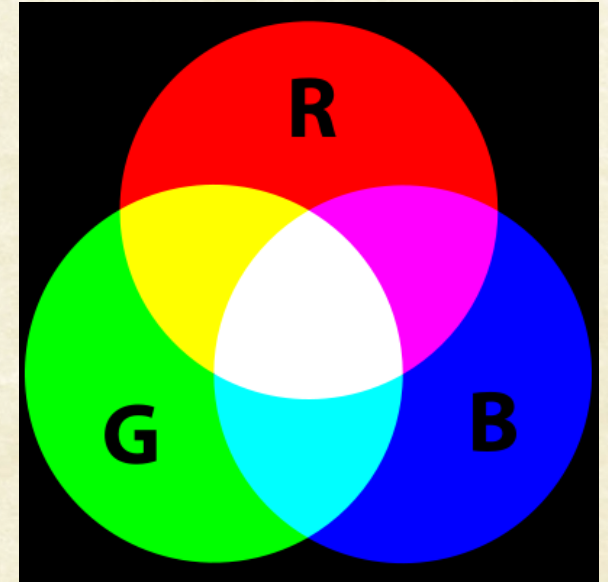
R,G,B Channels of an Image





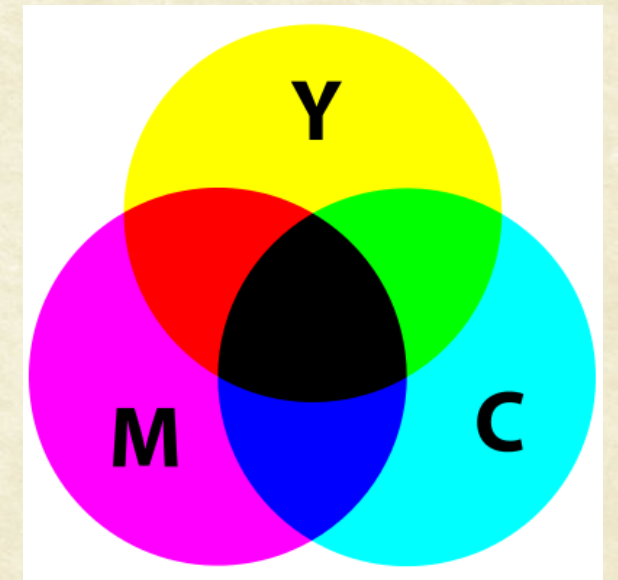
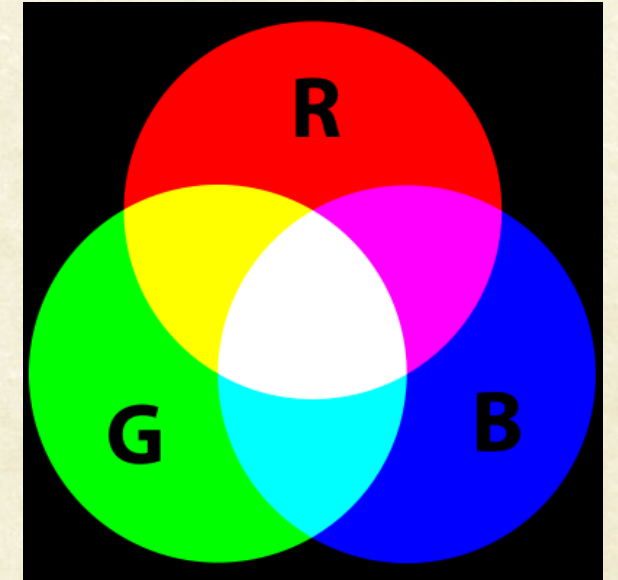
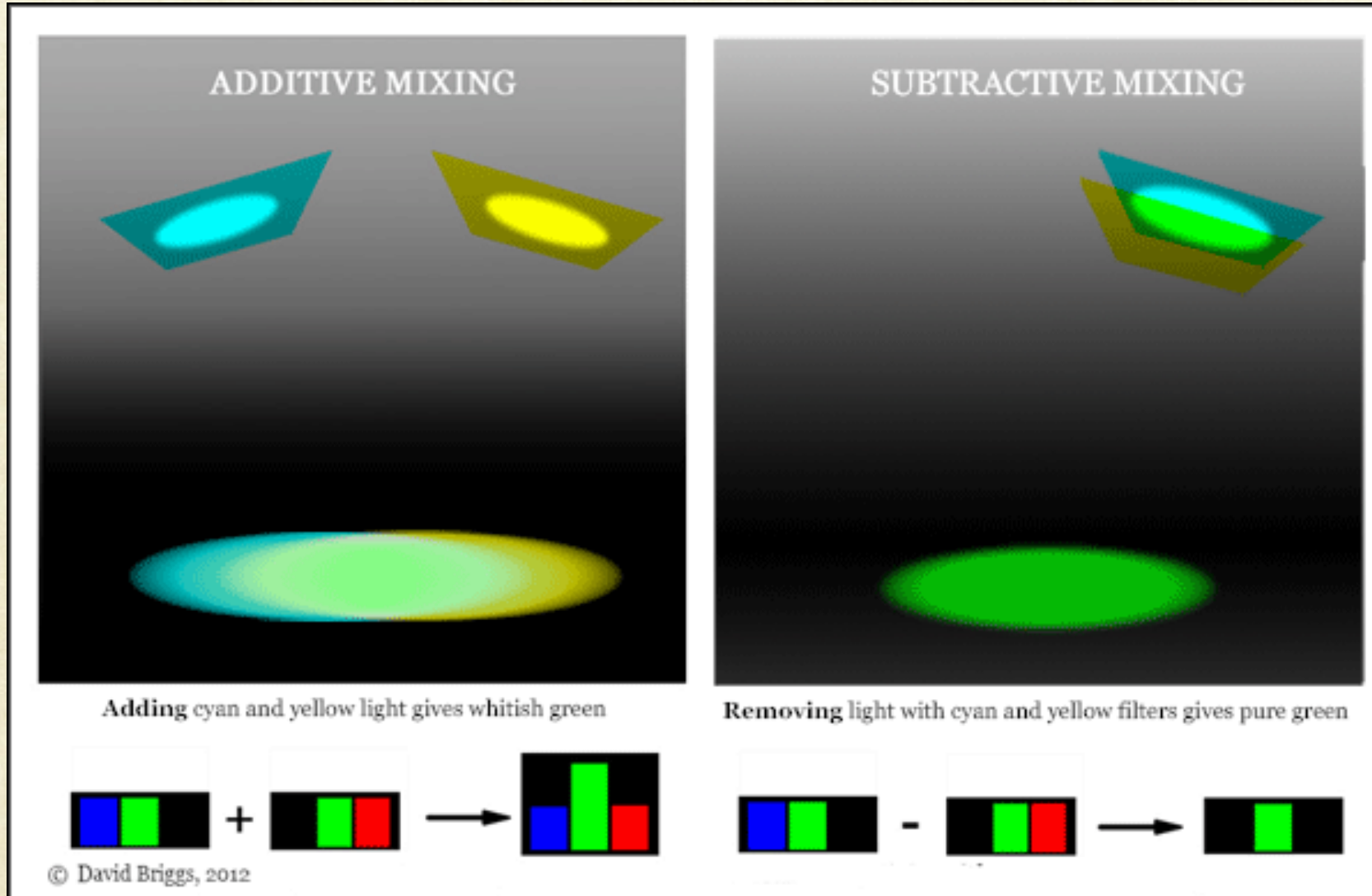
Primary Colors

- Subtractive (Painting, printing, etc.)
 - Prevent certain wavelengths from being reflected
 - Starting with white paper, each added colored ink “subtracts” available color from the starting medium.
 - For pure white , leave the paper unprinted (nothing can be whiter than the paper it is printed upon).



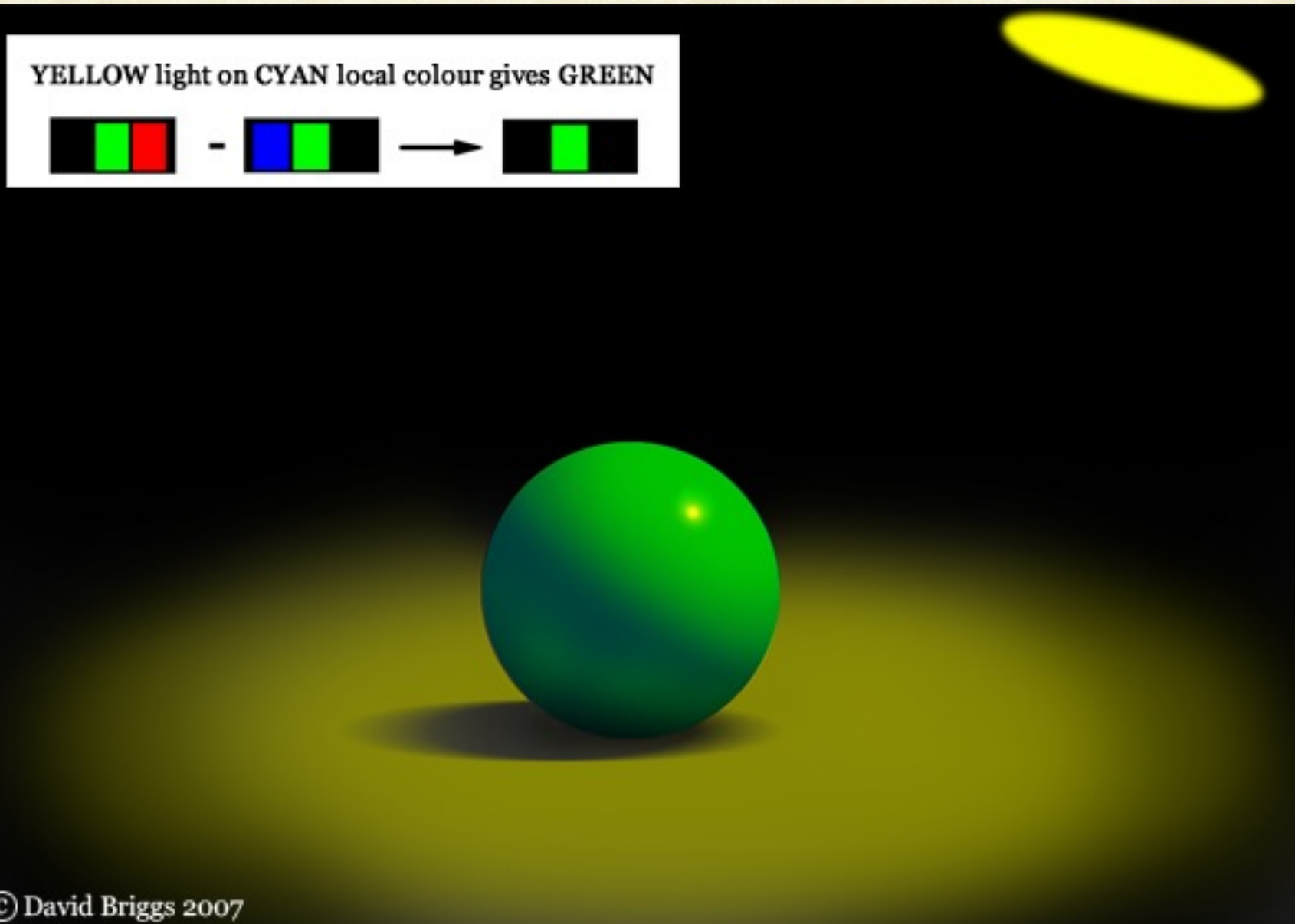


RGB vs CMY



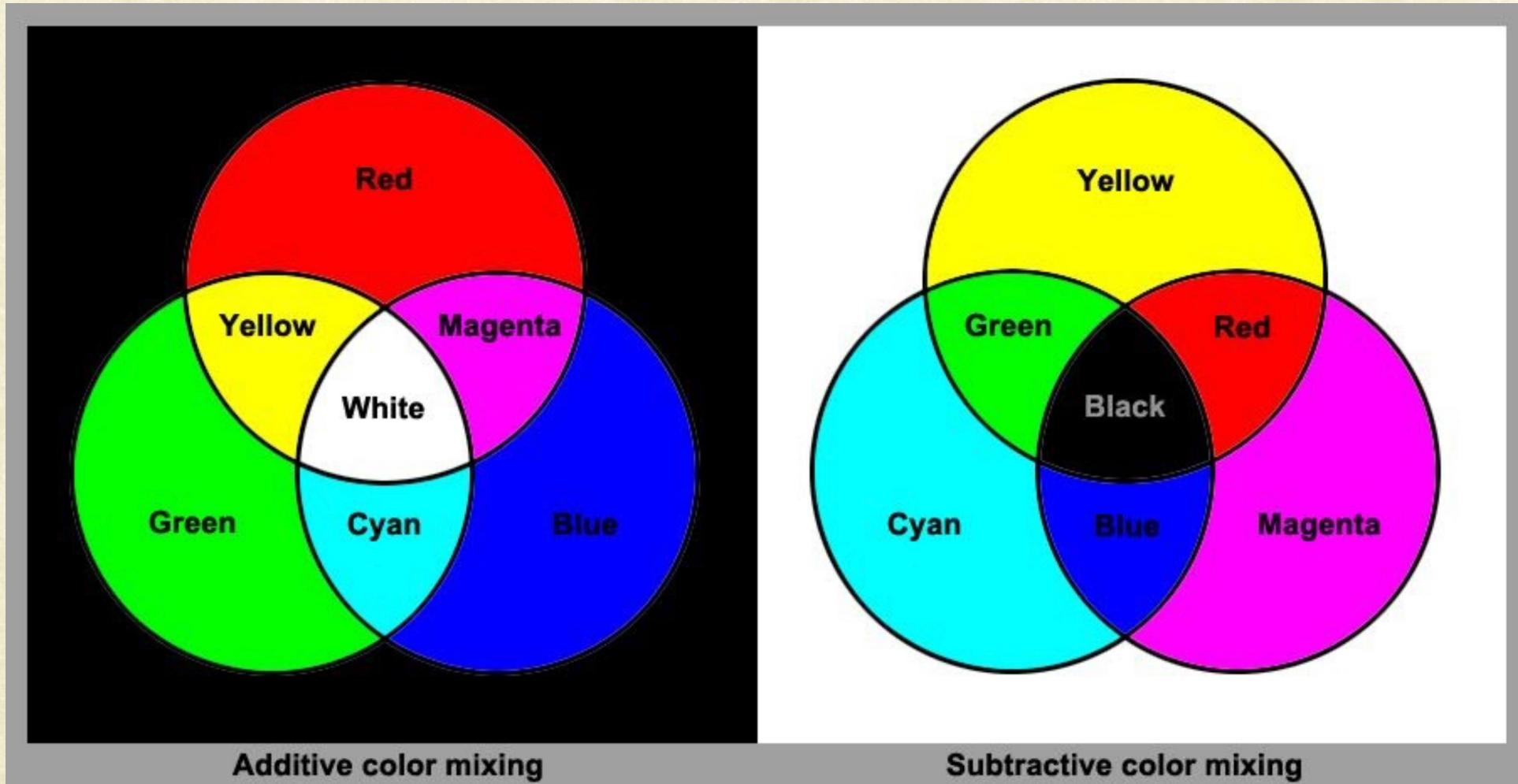


Objects in Colored Light





Additive vs. Subtractive color mixing



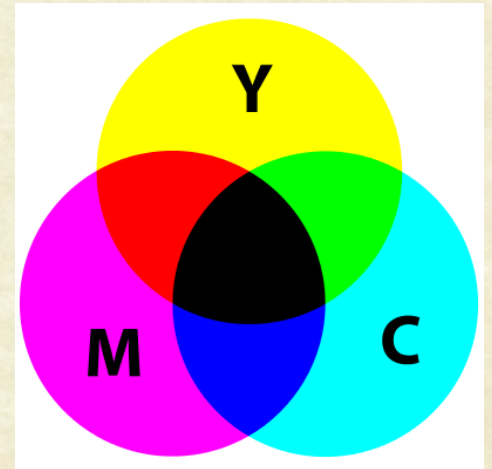
Magenta = Red + Blue
Cyan = Blue + Green
Yellow = Green + Red

Magenta = White - Green
Cyan = White - Red
Yellow = White - Blue



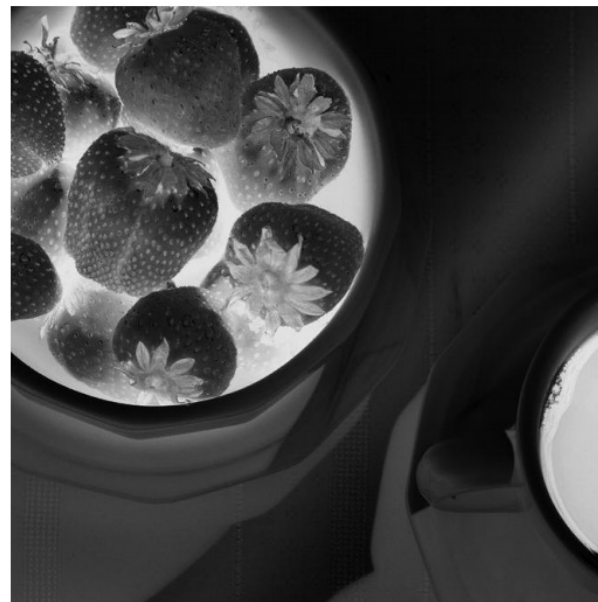
Why CMYK in Printing?

- Color pigments absorb specific wavelengths.
- CMY cover most lighter color ranges quite easily compared to RGB.
- BUT CMY by itself will not be able to create very deep dark colors (e.g. “true black,”)
- Black (designated “K” for “key color”) is added to CMY to achieve a much wider range of colors.

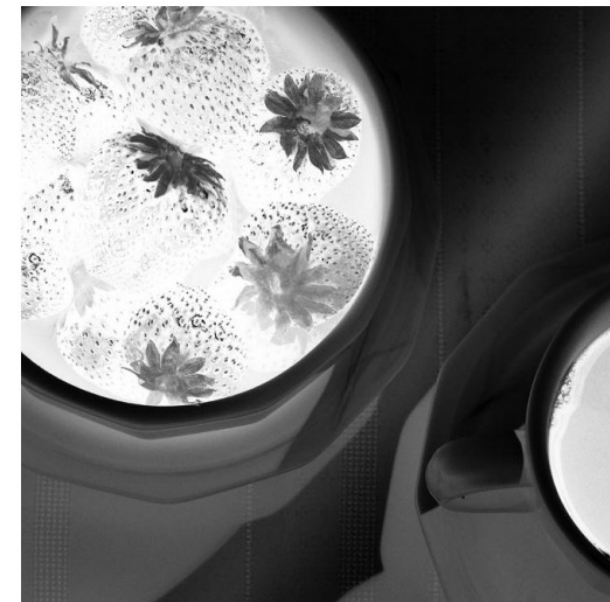




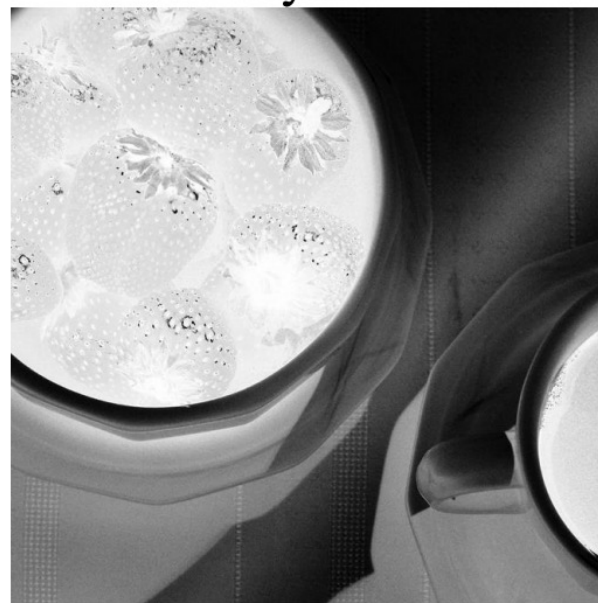
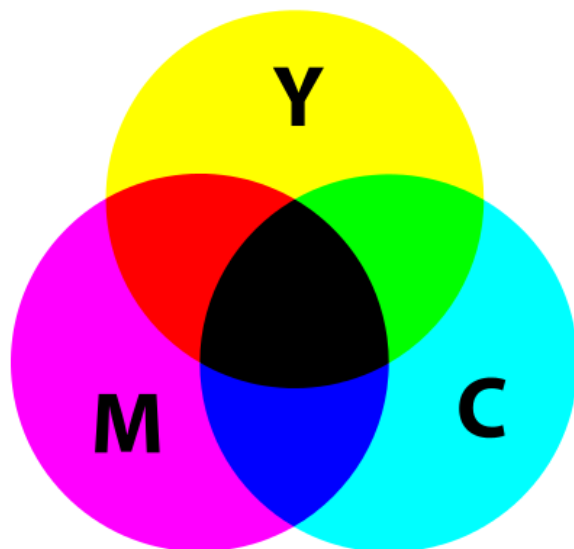
Full color



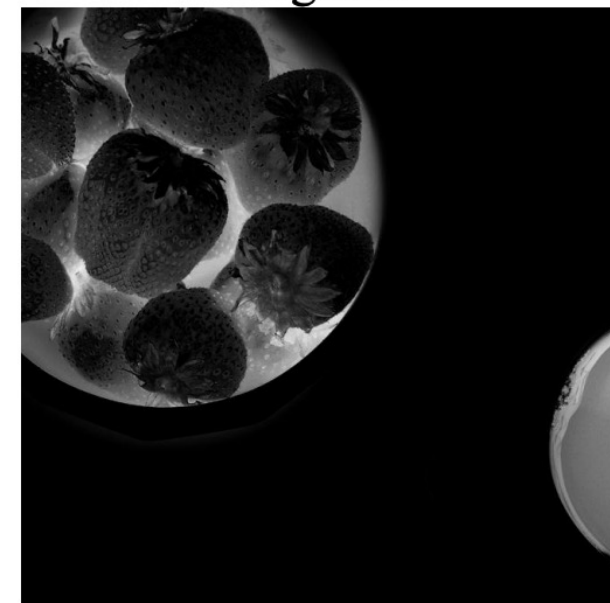
Cyan



Magenta



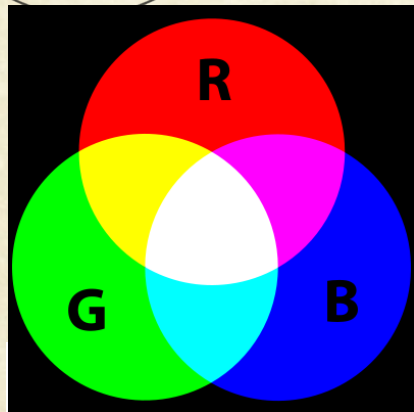
Yellow



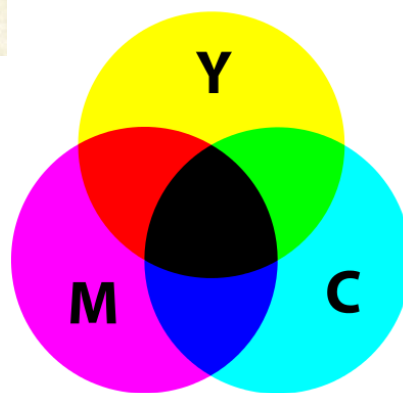
Black



RGB to CMY



$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



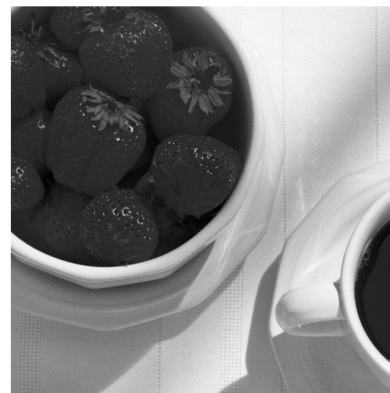
Full color



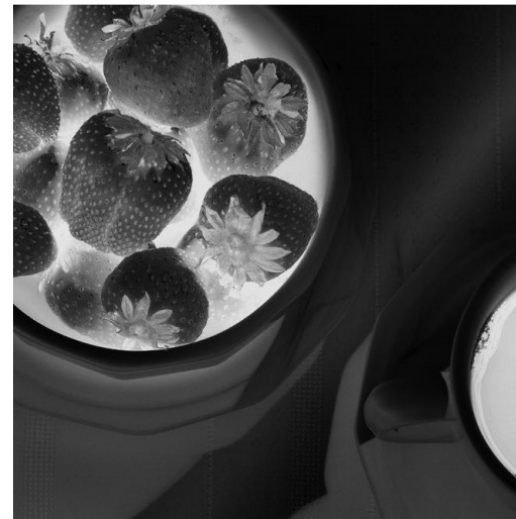
Red



Green



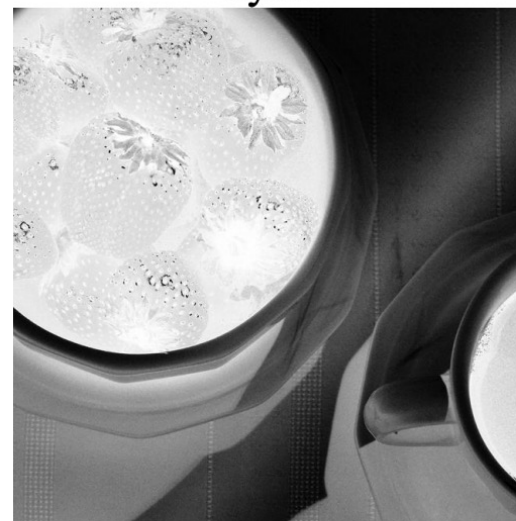
Blue



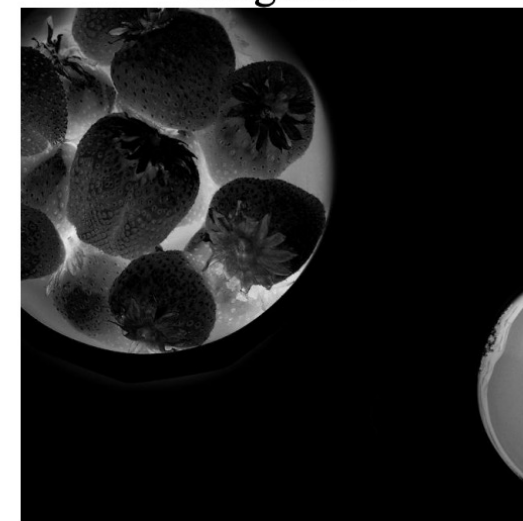
Cyan



Magenta



Yellow



Black



Additive and Subtractive Color Spaces

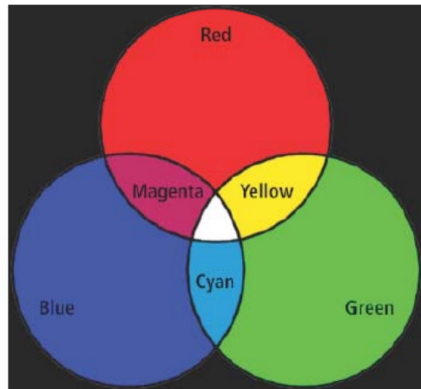


Original

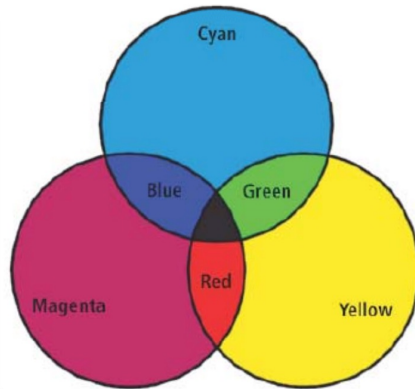
Red Band

Green Band

Blue Band



Magenta = Red + Blue
Cyan = Blue + Green
Yellow = Green + Red



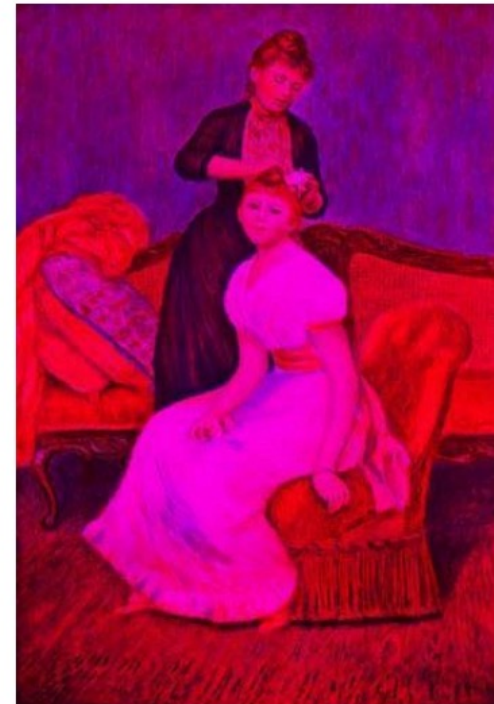
Magenta = White - Green
Cyan = White - Red
Yellow = White - Blue

Cyan



No Red

Magenta



No Green

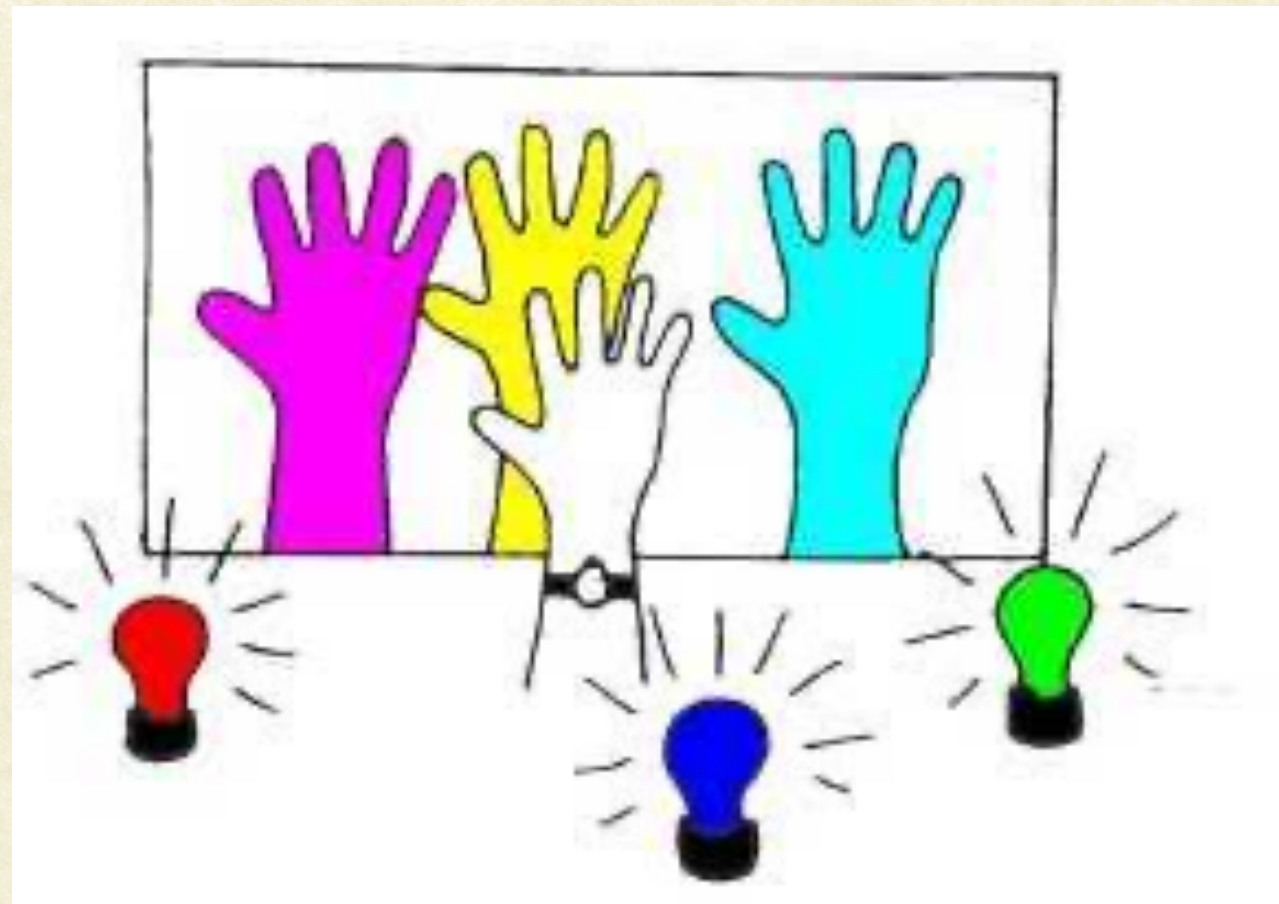
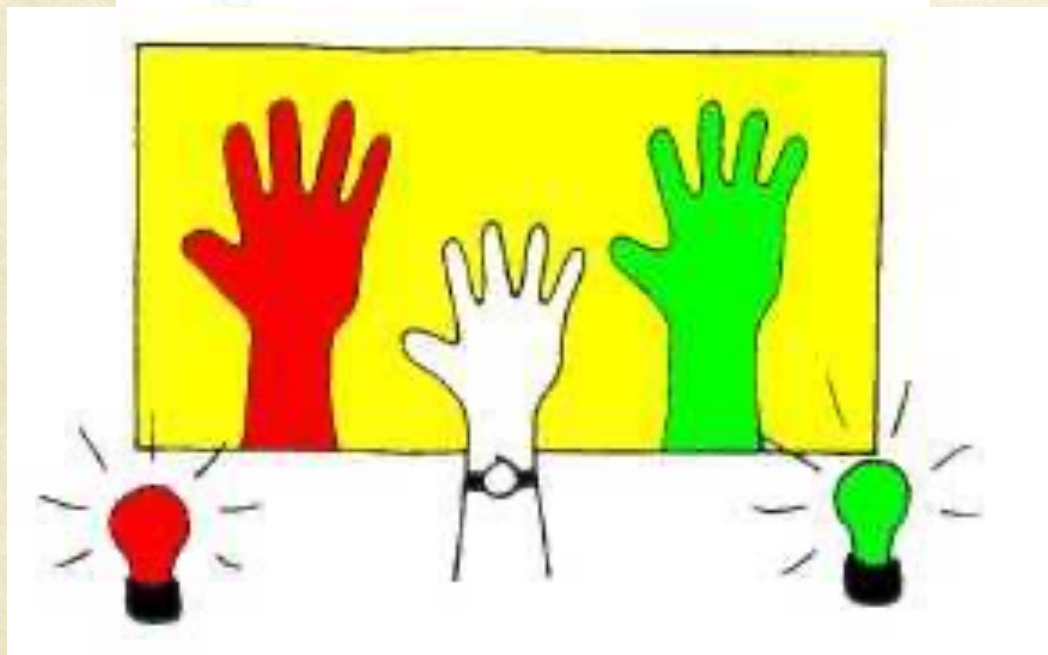
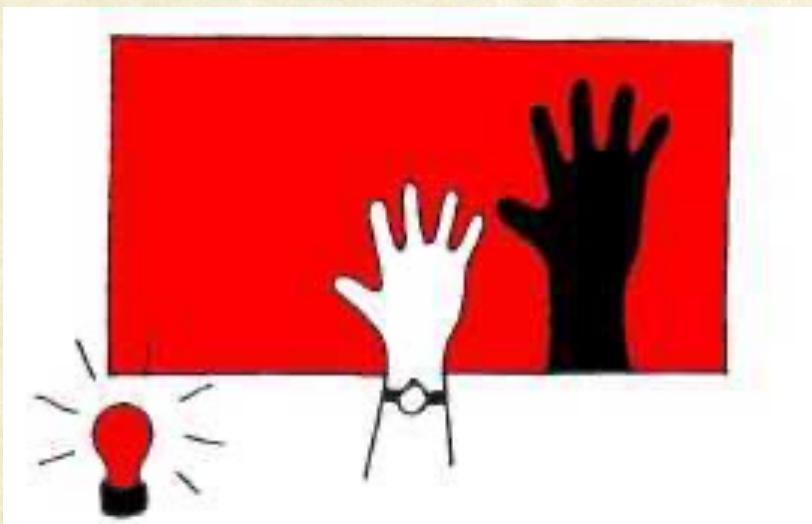
Yellow



No Blue



RGB vs. CMY





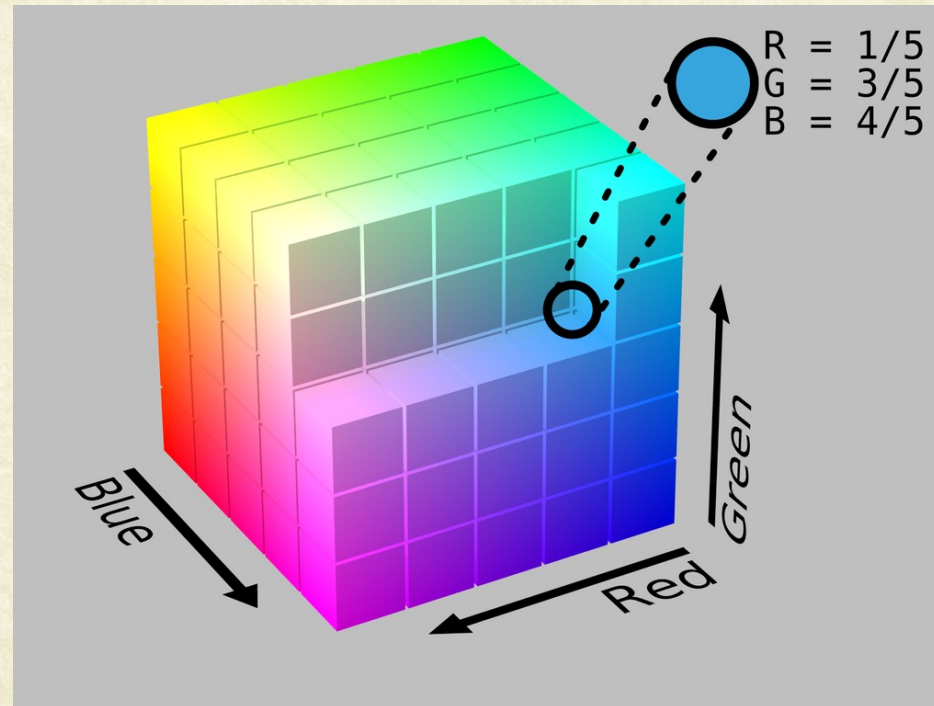
Color Models

- RGB
- HSI / HSY
- CIE LAB



RGB color space

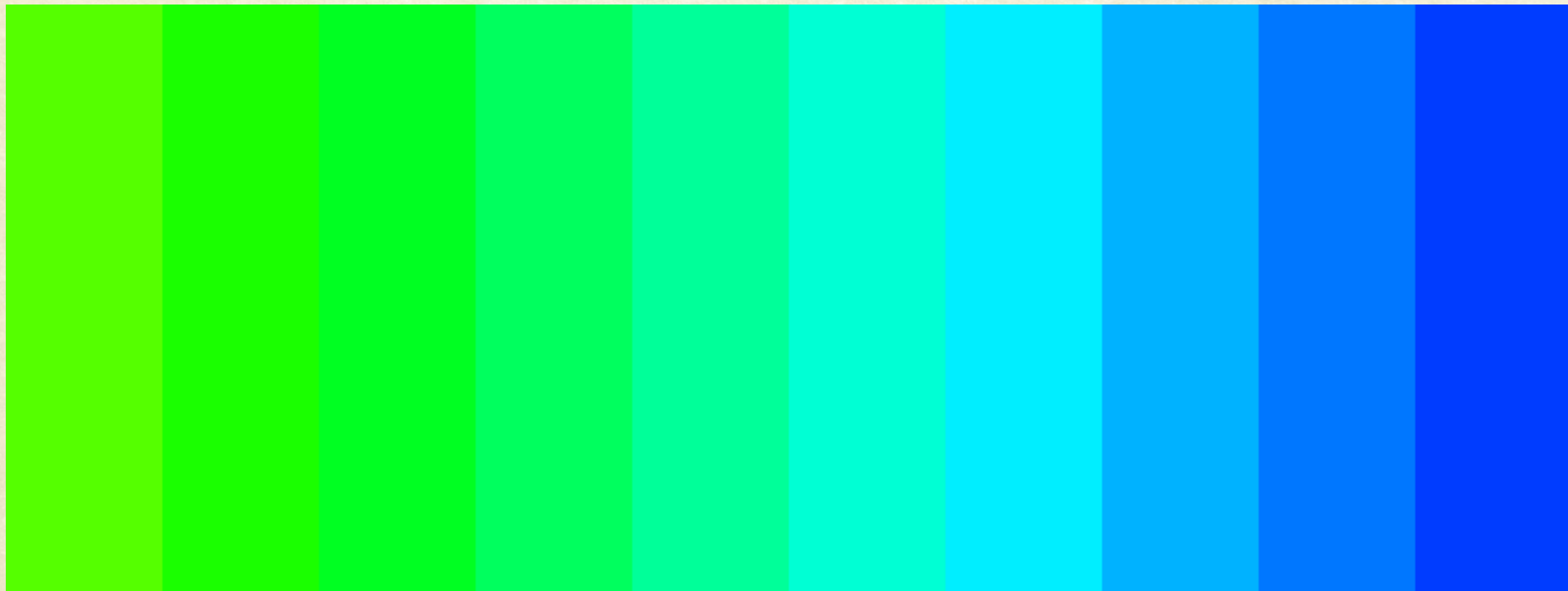
- Primary colors
- Additive color model $f(x, y) = \alpha_1 R + \alpha_2 G + \alpha_3 B$
- Perceptually non uniform



Courtesy: wikipedia



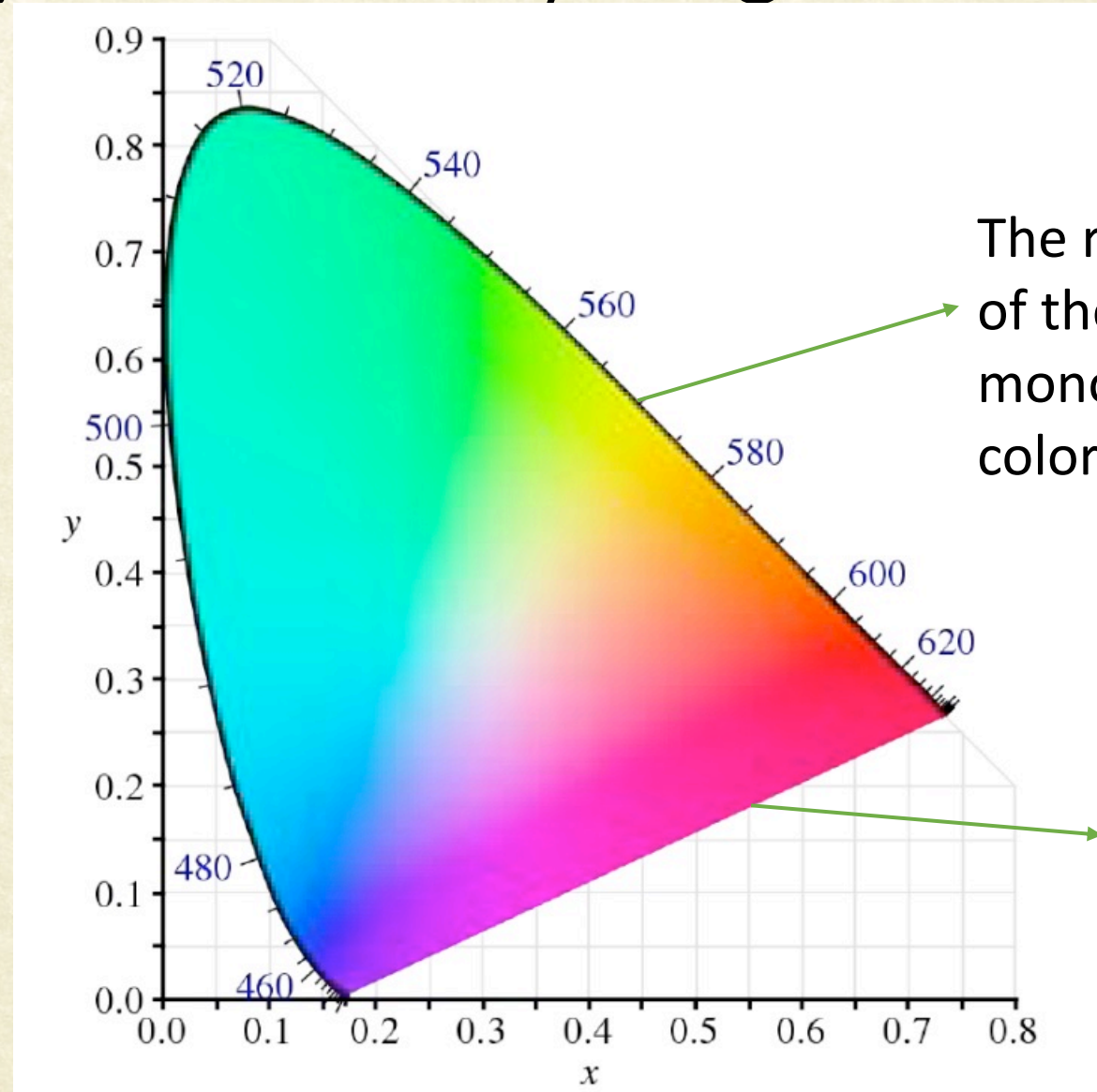
RGB : Non-Uniform Perceptual Space





CIE 1931 (x,y) chromaticity diagram

- Created by the International Commission on Illumination (CIE) in 1931
- Y x y: Luminance + the two most distinctive chrominance components
- Helps separate luminance and chromaticity
- A convenient representation for color values



The rim represents all of the pure monochromatic colors

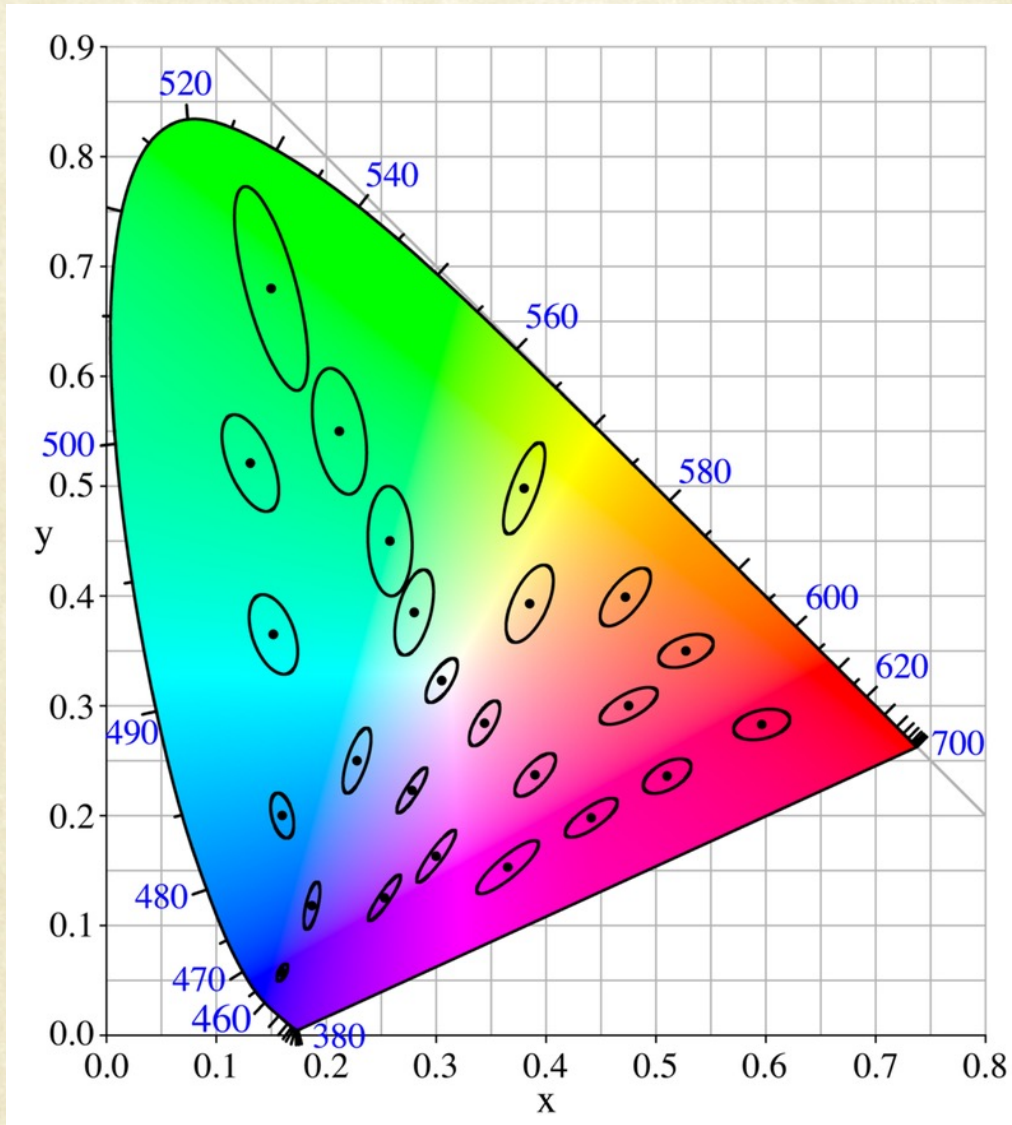
Purple Line

Visualizing CIE XYZ color space:

<https://www.youtube.com/watch?v=x0-qoXOCOow>



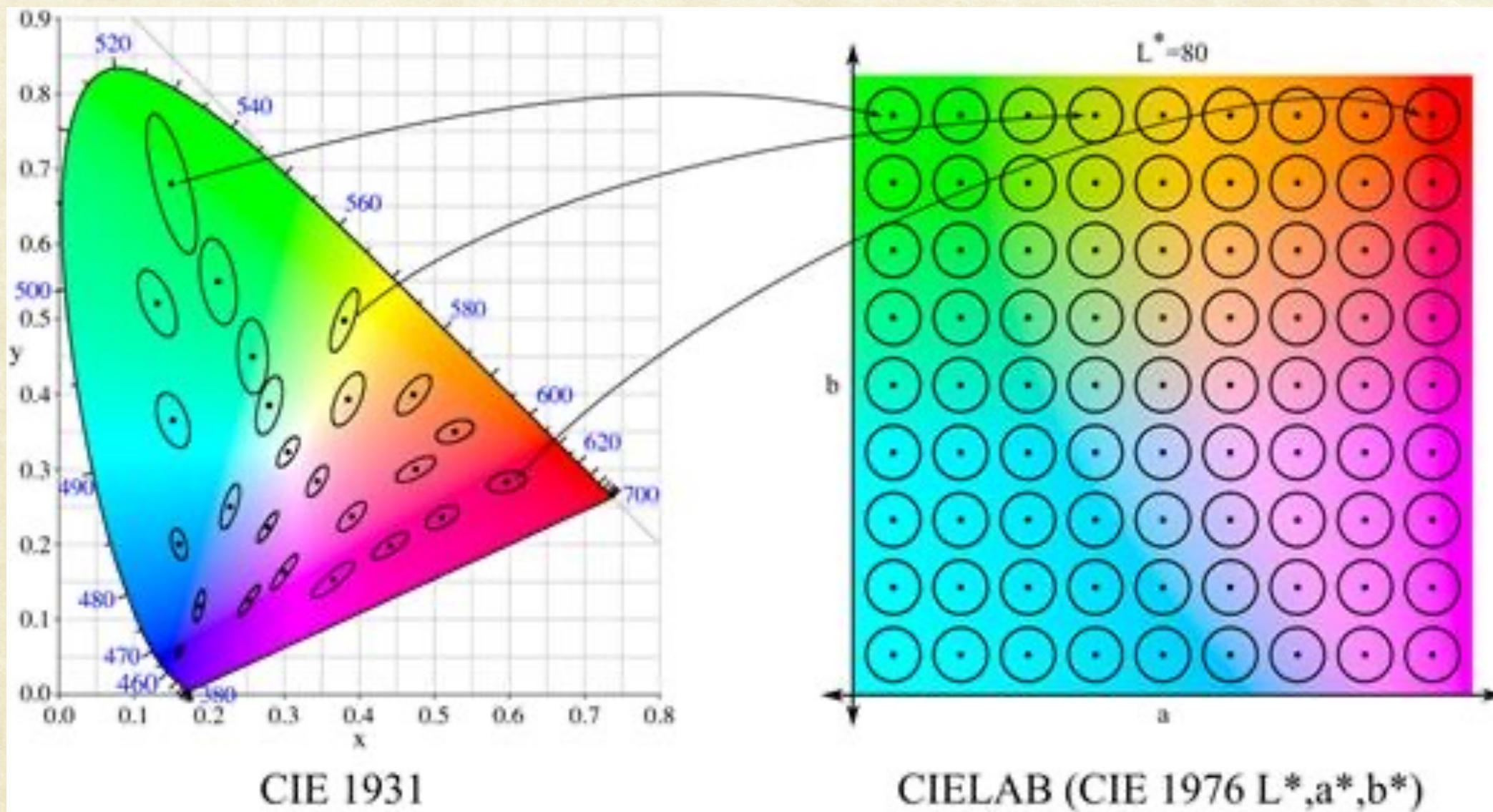
McAdam Ellipses



- **McAdam ellipses** refer to the region on a chromaticity diagram which contains all colors which are indistinguishable, to the average human eye, from the color at the center of the ellipse
- The contour of the ellipse represents the **just noticeable differences** of chromaticity



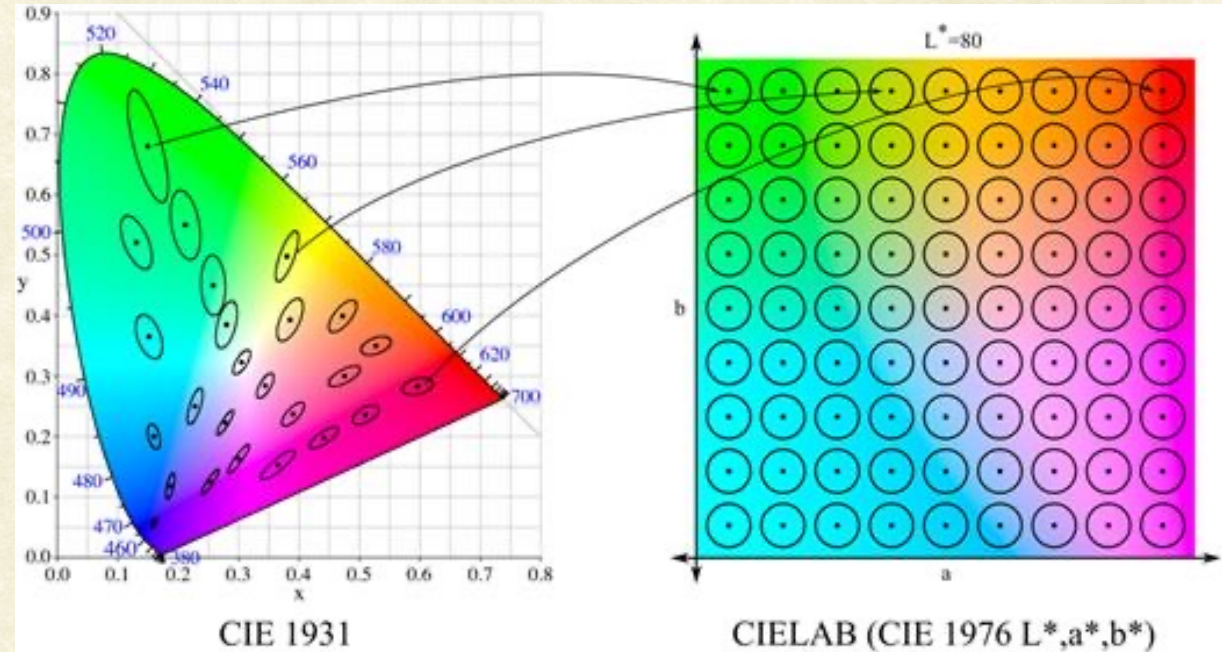
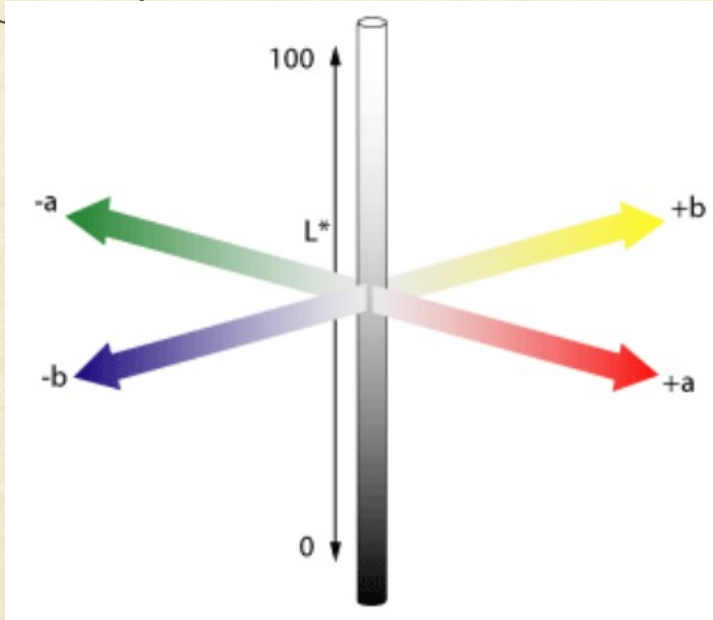
CIE Lab color space



Ideal scenario



CIE Lab color space



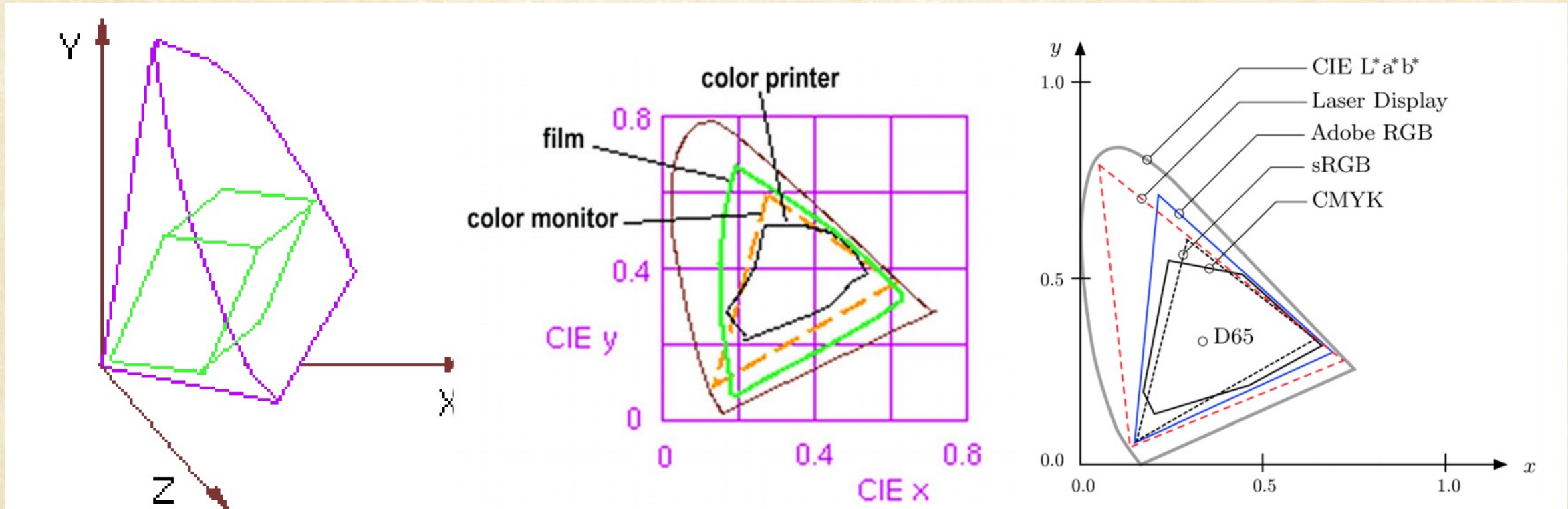
Components:

- Luminosity L^*
- Color components
 - a^* : color hue, saturation along green-red axis
 - b^* : color hue, saturation along blue-yellow axis
- Makes changes in CIE space linear with respect to human perception
- Popular in high-quality photographic applications



Color Gamuts of Devices

- The RGB color cube sits within the CIE color space
- CIE chromaticity diagram is useful to compare the gamut of various devices
 - e.g., compare color printer and monitor color gamuts.





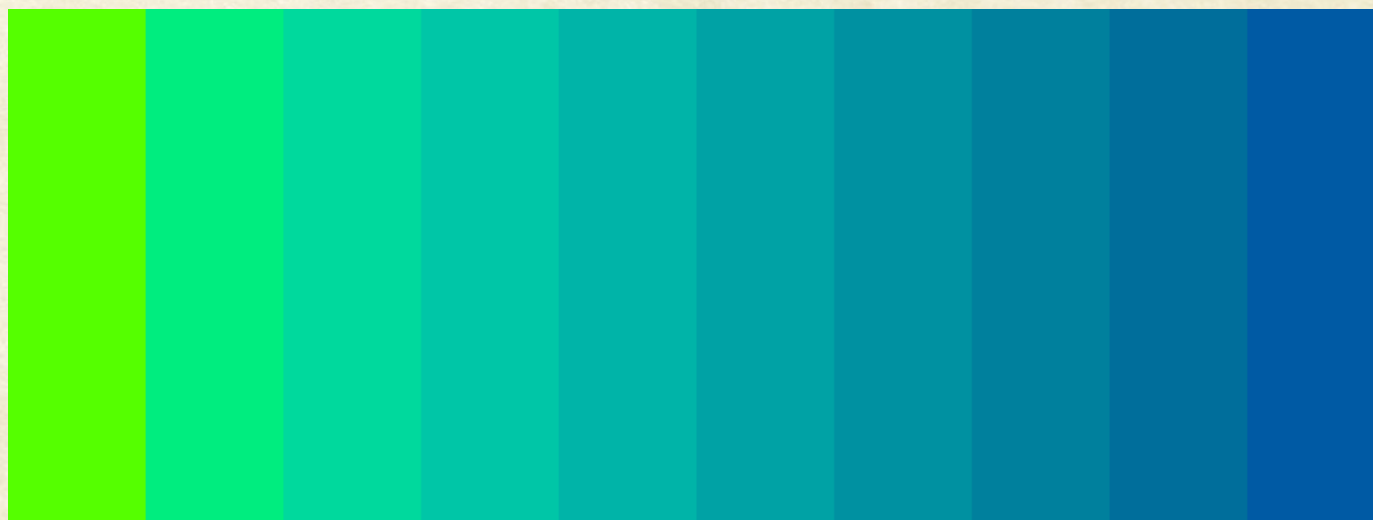
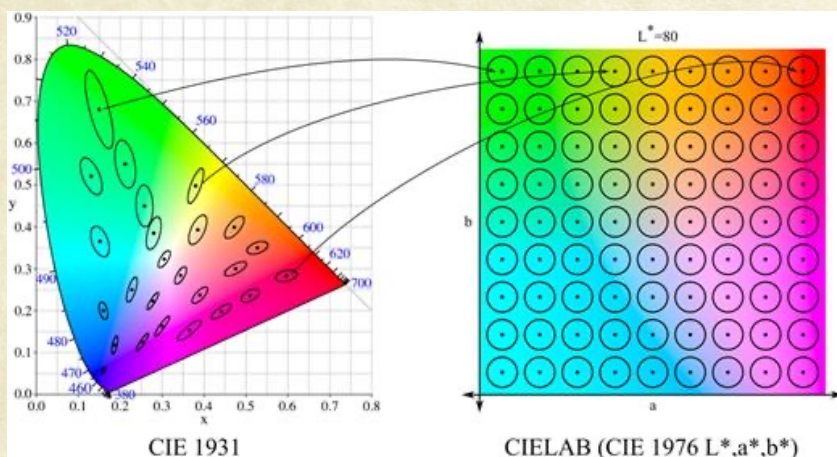
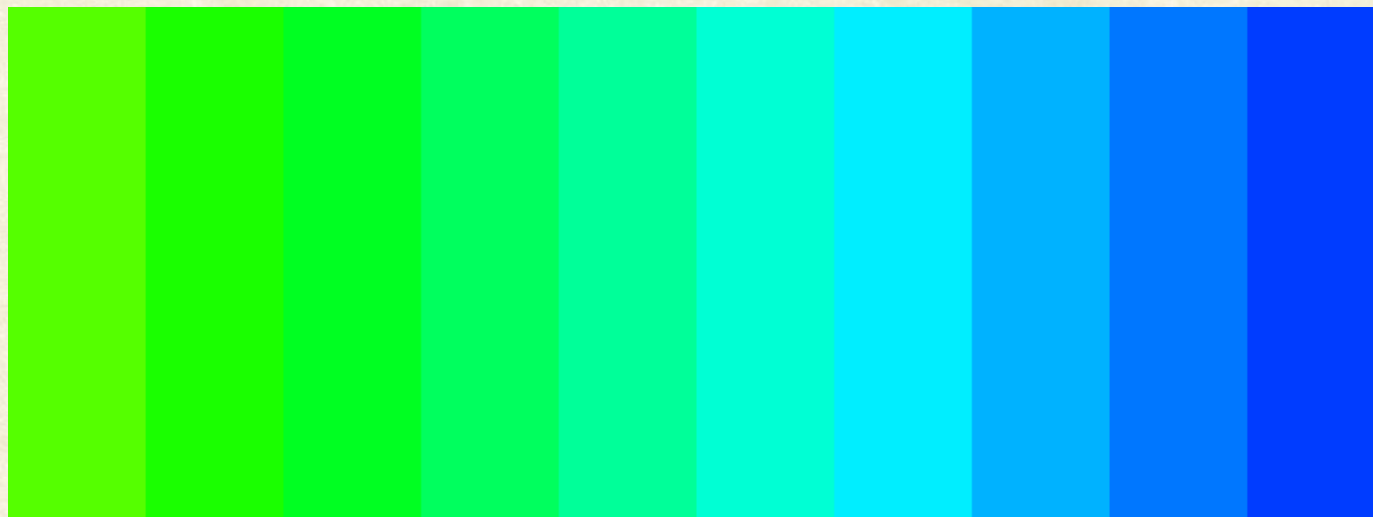
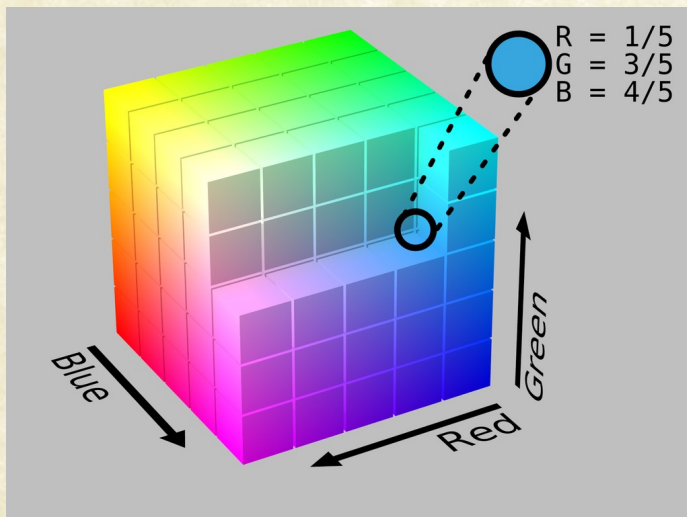
Measuring Color Differences

- Due to its uniformity with respect to human perception, the differences between colors in $L^*a^*b^*$ color space can be determined as **Euclidean distance**.

$$\begin{aligned}\text{ColorDist}_{\text{Lab}}(\mathbf{C}_1, \mathbf{C}_2) &= \|\mathbf{C}_1 - \mathbf{C}_2\| \\ &= \sqrt{(L_1^* - L_2^*)^2 + (a_1^* - a_2^*)^2 + (b_1^* - b_2^*)^2}\end{aligned}$$



Perceptually Uniform Color Spaces

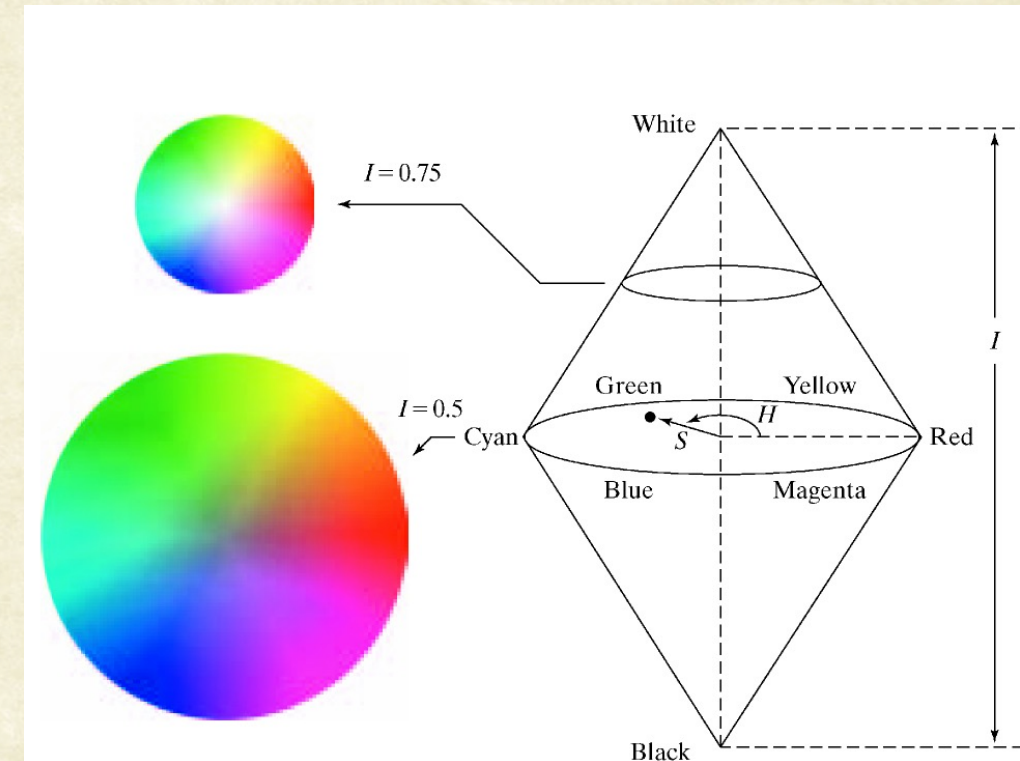


<https://matplotlib.org/3.1.1/tutorials/colors/colormaps.html>



HSI color space

- **Hue (H):**
 - Dominant colour perceived by an observer
 - When we call an object red or orange, we refer to its hue.
- **Saturation (S):**
 - Amount of white light mixed with a hue
 - Pure colors are fully saturated.
 - Pink (red+white) is less saturated.
- **Brightness (I)** : achromatic notion of intensity





HSI color space

■ HUE

- A subjective measure of color
- Average human eye can perceive ~200 different colors

■ Saturation

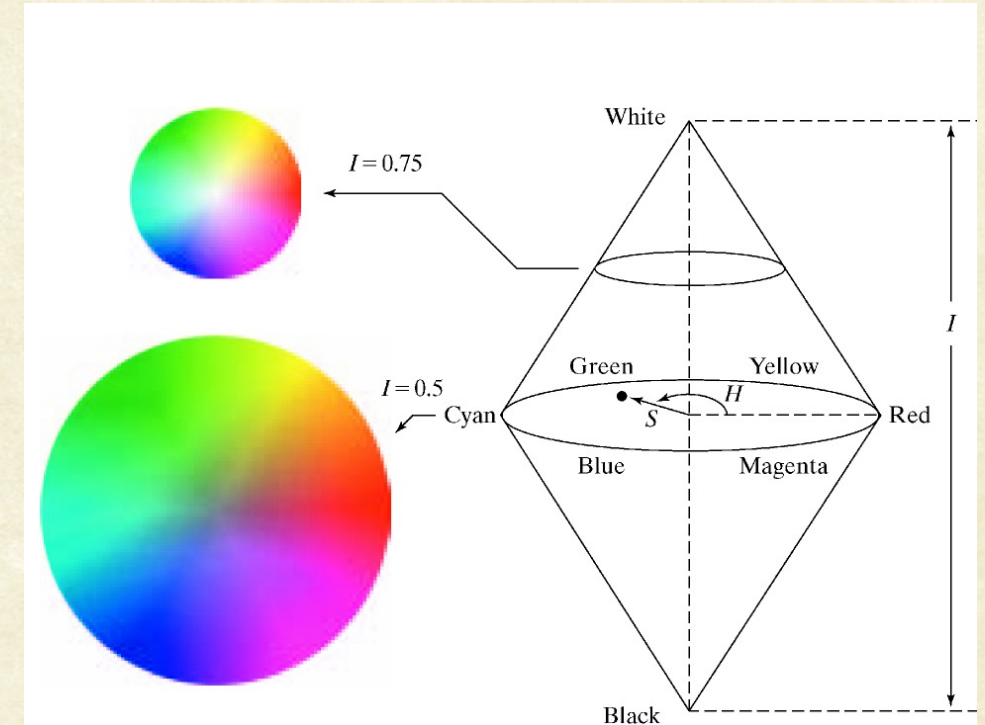
- Relative purity of the color. Mixing more “white” with color reduces its saturation.
- **Pink** has the same **hue** as **red** but less **saturation**

■ Intensity

- The brightness or darkness of an object

RGB : great for color generation

HSI : great for color description



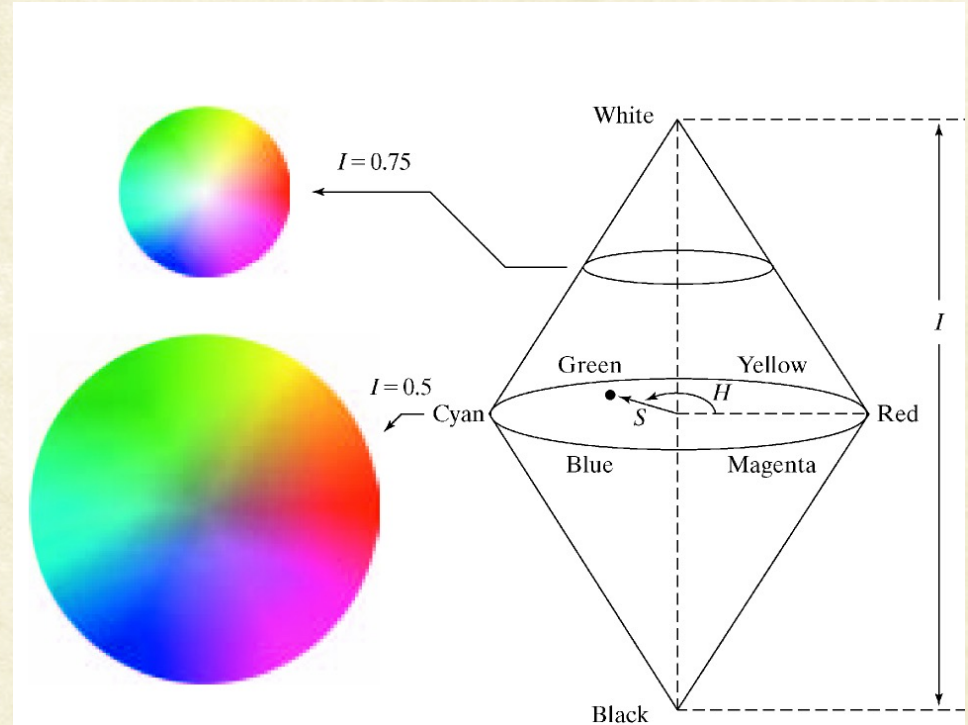


HSI color space

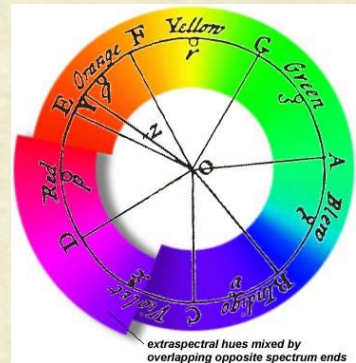
- **Hue** is defined as an angle
 - 0 degrees is **RED**
 - 120 degrees is **GREEN**
 - 240 degrees is **BLUE**

Saturation = % of distance from center : [0,1]

- **Intensity** is denoted as the distance “up” the axis from black.
 - Values range from 0 to 1

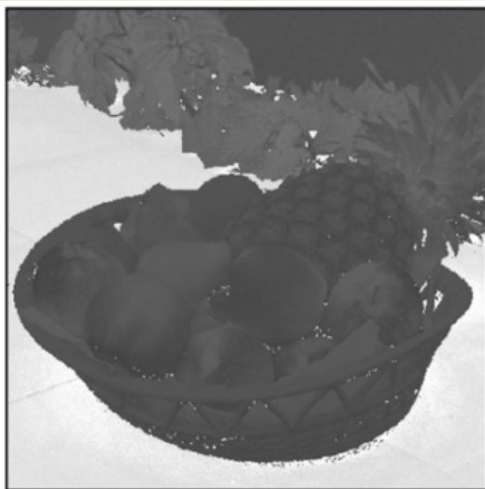


RGB : great for color generation
HSI : great for color description

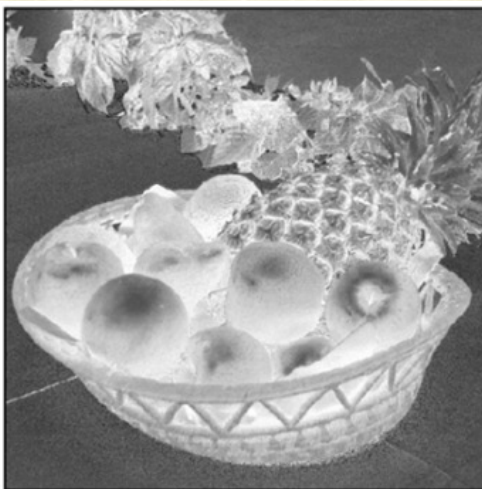




HSI vs Lab



H_{HSV}



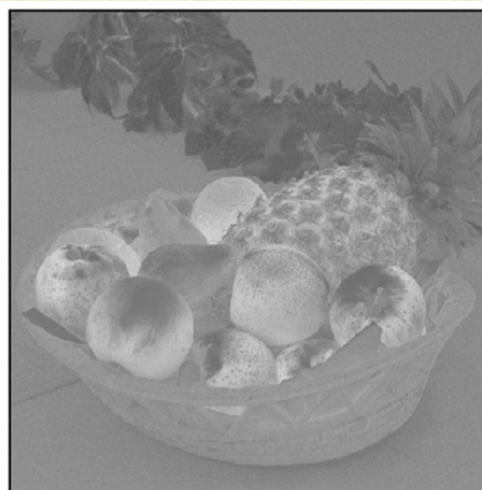
S_{HSV}



V_{HSV}



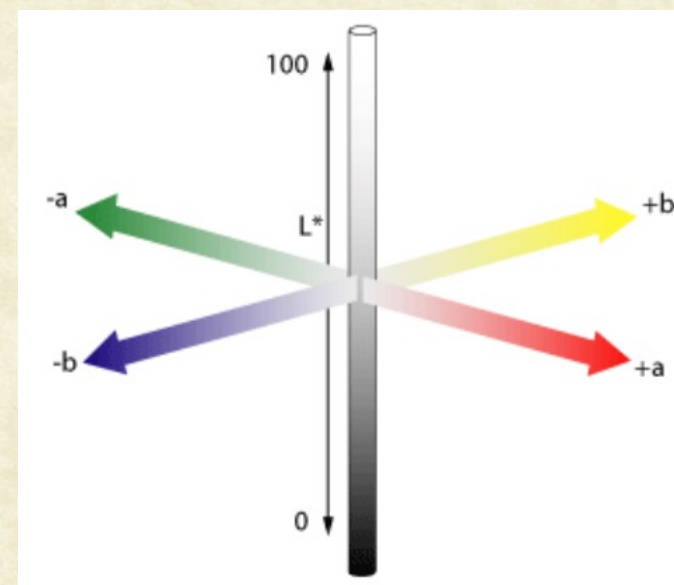
L^*



a^*



b^*



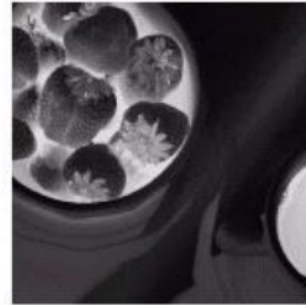


- CMY vs RGB vs HSI

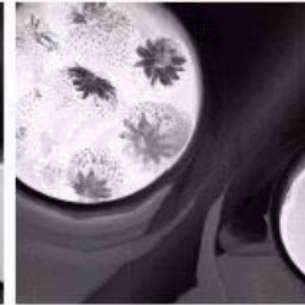


Full color

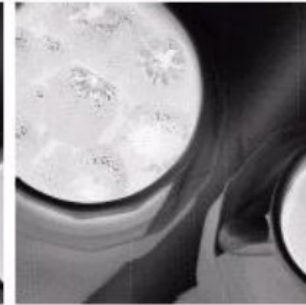
Full color image



Cyan



Magenta

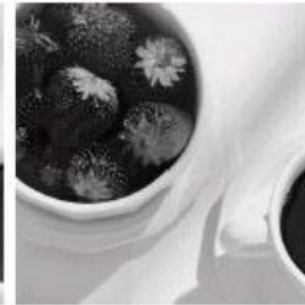


Yellow

C,M,Y components.



Red

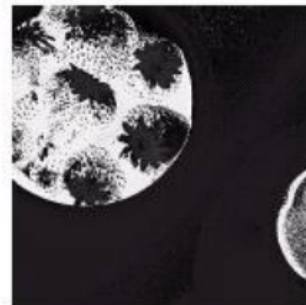


Green



Blue

R,G,B components.



Hue



Saturation



Intensity

H,S,I components.



Coloring Mughal-e-Azam

- The colourisation team spent 18 months developing software for colouring the frames, called "Effects Plus".
- Designed to accept only those colours whose hue would match the shade of grey present in the original film. This ensured that the colours added were as close to the real colour as possible.
- The authenticity of the colouring was later verified when a costume used in the film was retrieved from a warehouse, and its colours were found to closely match those in the film.
- Every shot was finally hand-corrected to perfect the look. The actual colourisation process took a further 10 months to complete. (cost: Rs. 2 to 10 cr)





Mayabazar (1957), colorized (2007)



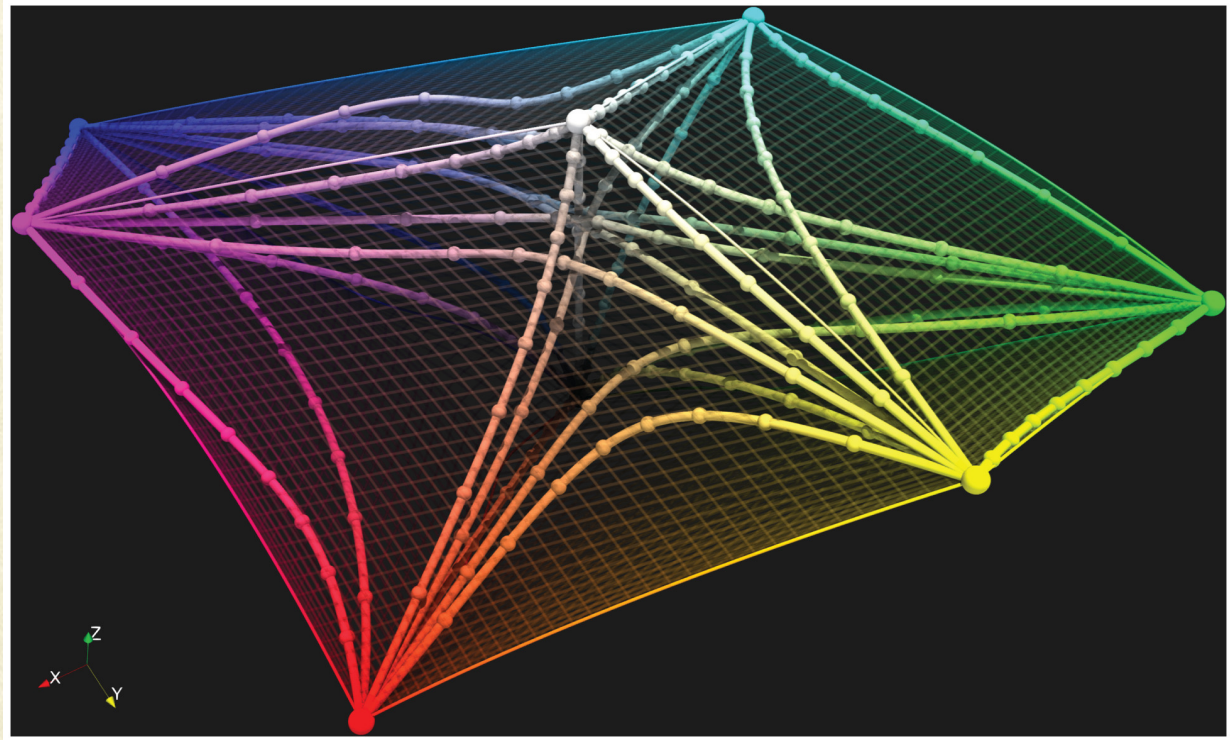
The Film Heritage Foundation announced in March 2015 that they would be restoring *Mayabazar*, along with a few other Indian films from 1931 to 1965, as a part of their restoration projects carried out in India and abroad in accordance with international parameters. **The foundation opposed digital colourisation**, stating that they "believe in the **original repair** as the way the master or the creator had seen it"



Errors in Perceptual Color Space

- Roxana Bujack, Emily Teti, Jonah Miller, and Terece L. Turton, “The non-Riemannian nature of perceptual color space”, Proceedings of the National Academy of Sciences, (April 2022), DOI: 10.1073/pnas.2119753119.
- Shows that the perceptual color space is probably non-Riemannian:

$$\Delta E(A, B) + \Delta E(B, C) > \Delta E(A, C)$$



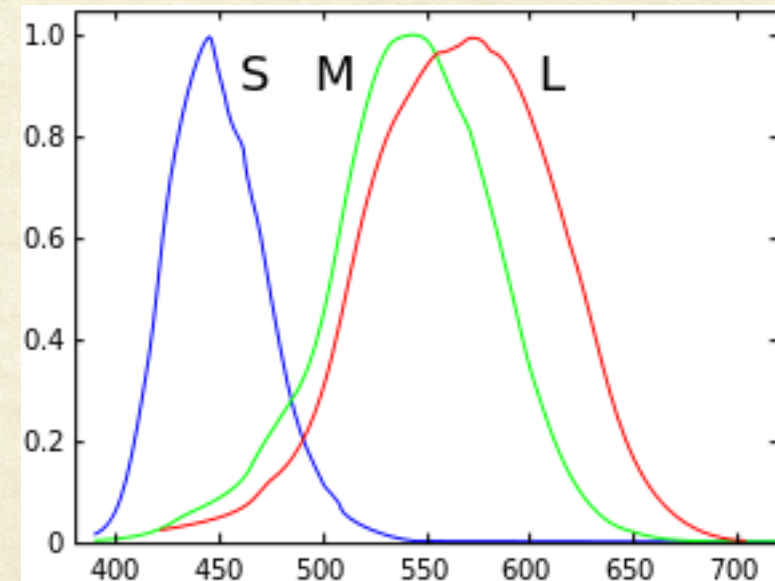
Geodesics in CIE Lab color space: Known since 2000



LMS color space

- Used when performing **chromatic adaptation** (estimating appearance of a sample under a different illuminant)
- Also useful in the study of color blindness, when one or more cone types are defective.

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 0.7328 & 0.4296 & -0.1624 \\ -0.7036 & 1.6975 & 0.0061 \\ 0.0030 & 0.0136 & 0.9834 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$





References

- “Colorimetry”, Ohta and Robertson, John Wiley and Sons Ltd
- <https://hypjudy.github.io/2017/03/19/dip-histogram-equalization/>
- http://www.inf.u-szeged.hu/ssip/2008/presentations2/Gordan_LectureSSIP08.pdf



Questions?